# Capping the Capacity Market: A Supply Function Equilibrium Analysis of Price Controls in PJM

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**Abstract**

In January 2025, Pennsylvania Governor Josh Shapiro negotiated a settlement reshaping PJM's capacity market demand curve: a price cap at $325/MW-day UCAP and a price floor at $175/MW-day UCAP for the 2026/27 and 2027/28 delivery years, projected to save consumers over $21 billion. We evaluate this intervention using a calibrated Supply Function Equilibrium model of PJM's Base Residual Auction. Under the observed market structure — three pivotal suppliers with residual supply indices below unity — equilibrium prices reach the VRR Point (a) cap in 2026/27 ($329/MW-day) and exceed the settlement cap by $4–$8/MW-day at the RTO level, with larger gaps in constrained Locational Deliverability Areas. The settlement is therefore binding but leaves the underlying markup structure nearly intact: the price-cost margin falls by less than one percentage point (from 0.544 at the SFE equilibrium to 0.538 at the settlement cap). The $21 billion savings figure uses the unconstrained VRR Point (a) price as the counterfactual; our structural model provides an alternative estimate of approximately $612 million for the two-year period — 35 times smaller — because the strategic equilibrium falls near the design ceiling rather than far above the settlement cap. The complaint's lead-time argument is partial: even at short lead times, prices inform retirement decisions, mobile resource deployment, contracting, and investment

expectations. Transmission expansion (increasing the Capacity Emergency Transfer Limit) reduces equilibrium prices organically by expanding the competitive fringe, preserving the scarcity signal while addressing the structural source of high prices.

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# 1 Introduction

On December 30, 2024, the Commonwealth of Pennsylvania filed a complaint with the Federal Energy Regulatory Commission alleging that PJM's capacity auction demand curve was miscalibrated: its price cap was set too high given near-term expectations for new entry. The complaint cited PJM's July 2024 Base Residual Auction for the 2025/26 delivery year, which cleared at \$269.92/MW-day — an increase of more than 800 percent from the prior year — as evidence that the existing market design imposed unnecessary costs on 65 million consumers across 13 states (Office of Governor Josh Shapiro, 2025). Six weeks later, on January 28, 2025, Governor Josh Shapiro announced an agreement with PJM: the demand curves for the 2026/27 and 2027/28 auctions would be modified to impose a price cap of approximately \$325/MW-day (unforced capacity, UCAP) and a price floor of approximately \$175/MW-day UCAP, applied to both the RTO and all constrained Locational Deliverability Areas (Bresler and Keech, 2025). The agreement was projected to save consumers over \$21 billion over the two delivery years.

The settlement creates a two-tier price ceiling in PJM's capacity market. PJM's existing Variable Resource Requirement (VRR) curve already contains a design-level cap at Point (a), set at approximately 1.5 times the Net Cost of New Entry — an investment-adequacy ceiling calibrated to the annualized cost of a reference generating unit and intended to prevent prices from permanently exceeding the entry threshold. The settlement imposes a second, lower cap at $\bar{p}_S \approx$ \$325/MW-day, below the Point (a) level, while simultaneously adding a floor at $\bar{p}_f \approx$ \$175/MW-day that clears all capacity offered at or below that level. The mechanism is implemented by reshaping the demand curve itself: a flat top at $\bar{p}_S$, the existing sloped segment between $\bar{p}_S$ and $\bar{p}_f$, and a flat bottom at $\bar{p}_f$ (Bresler and Keech, 2025). This paper asks: what price would PJM's 2026/27 and 2027/28 auctions have cleared at absent the settlement modification, and what are the consequences of suppressing prices to $\bar{p}_S$?

We answer the question using a calibrated Supply Function Equilibrium (SFE) model of the PJM Base Residual Auction. In the model, $K$ symmetric strategic sellers simultaneously

submit supply functions against PJM's VRR demand curve, which is publicly known before bidding. Following Klemperer and Meyer (1989) and Green and Newbery (1992), equilibrium is characterized by a first-order ordinary differential equation (ODE) system in the sellers' supply functions, integrated backward from the capacity-constrained boundary. The equilibrium is unique because the binding-capacity conditions of Holmberg (2008) are satisfied: PJM's reliability requirement implies positive probability that aggregate supply reaches the capacity ceiling at the VRR Point (a) cap, pinning the arbitrary constant in the ODE. We calibrate the model to three benchmark delivery years — 2023/24 (low price, old VRR), 2025/26 (price spike, old VRR), and 2026/27 (at-cap, new VRR) — using published PJM planning parameters, BRA clearing results, and IMM State of the Market cost data. The SFE equilibrium price $p^*$ is the counterfactual: where the unmodified VRR curve would have cleared under strategic bidding. This counterfactual is what is needed to evaluate the settlement's claim of \$21 billion in savings.

Four findings emerge. First, under the observed market structure — $K = 3$ pivotal suppliers with residual supply indices below unity — SFE equilibrium prices reach the VRR Point (a) cap in 2026/27 and exceed the settlement cap $\bar{p}_S = \$325$/MW-day. Price-cost margins (Lerner indices) are 0.54–0.65 across benchmark years. The at-cap 2026/27 outcome is consistent with $K \leq 3$: four or more symmetric sellers would predict interior clearing below the cap, so the observed outcome establishes $K \leq 3$ as the relevant structural regime. Second, the settlement is binding. The SFE equilibrium $p^* = \$329$/MW-day at the RTO level and substantially higher in constrained LDAs (where 2025/26 prices reached \$444–\$466/MW-day); in both cases, equilibrium prices exceed $\bar{p}_S = \$325$/MW-day. Suppressing prices from $p^*$ to $\bar{p}_S$ distorts entry incentives and destroys information about scarcity conditions (Hayek, 1945). Third, the settlement's \$21 billion savings claim uses the unconstrained VRR Point (a) price (approximately \$500/MW-day) as the counterfactual; our SFE model provides a structural alternative estimate that is lower but still exceeds the settlement cap. The difference between these counterfactuals is economically consequential: the Point (a) counterfactual includes

scarcity rents that are part of the market design, while the SFE counterfactual isolates strategic markup above competitive levels. Fourth, transmission expansion — modeled as increases in the Capacity Emergency Transfer Limit (CETL) — organically reduces equilibrium prices by shifting LDAs from cap-binding to interior-clearing regimes, addressing the structural source of high prices without suppressing price signals (Weitzman, 1974).

The remainder of the paper is organized as follows. Section 2 surveys the supply function equilibrium literature, the capacity market design literature, and the economics of price controls in oligopolistic markets. Section 3 reviews PJM's institutional context, including the VRR mechanism, auction irregularities, and the Shapiro settlement. Section 4 develops the theoretical SFE model. Section 5 describes data sources and calibration. Section 6 presents SFE equilibrium prices and comparative statics relative to the settlement cap. Section 7 analyzes the allocative consequences of the settlement's price band for entry, exit, and information. Section 8 evaluates the lead-time argument underlying the complaint. Section 9 examines the $21 billion savings claim. Section 10 considers structural alternatives to price caps. Section 11 concludes.

# 2 Literature Review

## 2.1 Supply Function Equilibria: Theory

The supply function equilibrium concept, introduced by Klemperer and Meyer (1989), considers an oligopoly in which each firm commits to a schedule mapping prices to quantities before demand is realized. Unlike Bertrand competition (commitment to a fixed price) or Cournot competition (commitment to a fixed quantity), the supply function strategy allows each firm to adapt its output to whatever market price emerges after the demand shock is realized. Klemperer and Meyer characterize the symmetric Nash equilibria of this game under additive demand uncertainty, showing that equilibria are characterized by a first-order differential equation whose solutions form a continuum: in general, any supply

function tracing a trajectory between the competitive schedule (price equals marginal cost) and the Cournot schedule (as if rivals bid at capacity) constitutes a Nash equilibrium. The specific equilibrium realized depends on market coordination, and demand uncertainty is what disciplines firms to choose upward-sloping supply functions over fixed prices.

Green and Newbery (1992) apply the Klemperer-Meyer framework to the England and Wales electricity spot market after privatization, modeling two dominant generators as strategic suppliers to a pool settled at a uniform price. Their calibrated duopoly shows that the Nash equilibrium in supply functions implies substantial markups above marginal cost—with prices near the top of the feasible equilibrium range—and that entry of additional firms (or divestiture into five units) would significantly reduce those markups. Green and Newbery establish the methodological template followed here: express demand as a known function of price, derive the symmetric first-order ODE, and integrate backward from the capacity-constrained boundary to obtain the equilibrium supply function. Their paper is the closest methodological antecedent to the present work, with the key difference that our demand function is PJM's explicit piecewise-linear VRR curve rather than a load- duration curve, and our market clears annually in a single forward auction rather than half-hourly in a spot pool.

The multiplicity problem endemic to SFE is resolved by Holmberg (2008), who shows that under conditions natural to capacity-constrained electricity markets—symmetric producers, inelastic demand, a price cap, and capacity constraints that bind with positive probability—the supply function equilibrium is unique. The key insight is that when rivals' capacity constraints are simultaneously binding, firm $i$ faces perfectly inelastic residual demand; the monopoly optimum at that moment is to charge exactly the price cap. This ties the boundary condition of the ODE: each firm's supply function must equal its full installed capacity at the price cap. The arbitrary constant in the ODE solution is thereby pinned, selecting a unique trajectory. This result is central to the identification strategy of the present paper, since the VRR Point (a) price serves as the price cap and the at-cap clearing outcomes in 2026/27 and

2027/28 are precisely the event Holmberg requires to be positive probability. Holmberg (2009) extends the uniqueness analysis to pay-as-bid auctions, showing that a unique symmetric equilibrium obtains there as well. Vasin et al. (2016) confirm existence and characterize additional properties of symmetric SFE in uniform-price settings with oligopolistic supply.

The effect of forward commitments on equilibrium supply functions is examined by Allaz and Vila (1993), who show in a Cournot setting that forward markets reduce equilibrium markups through a Stackelberg-style commitment effect. Anderson and Xu (2005) extend the SFE framework to settings with both forward contracts and price caps, establishing existence conditions for equilibria under these combined constraints. Capacity commitments in the BRA are formally analogous to forward sales, so the static SFE applied to the auction game incorporates these disciplining effects into the equilibrium concept.

## 2.2 Numerical Methods for SFE

The numerical computation of SFE relies on the differential equation characterization of equilibrium supply functions. For the symmetric case the ODE reduces to a single equation integrable backward from the capacity boundary. For asymmetric firms the system of $K$ coupled ODEs requires iterative methods. Anderson and Hu (2008) develop the leading algorithm for the asymmetric case: initialize all firms at competitive supply functions, solve each firm's single-firm ODE taking rivals' current functions as fixed, update simultaneously, and iterate to convergence. They establish that equilibrium solutions can be ordered—a partial ordering that guides the search for the strong equilibrium in which all but one firm are at capacity at peak demand—and demonstrate convergence using GAMS implementations for calibrated electricity market examples. The Anderson–Hu algorithm is the primary computational reference for the asymmetric extension described in Section 5.3.

Baldick et al. (2004) develop the analytically tractable special case in which both the demand curve and firms' marginal cost functions are linear. Under these restrictions the symmetric ODE has a closed-form solution parameterized by a single scalar, yielding explicit

comparative statics with respect to the number of firms and the demand slope. Linear SFE provides a useful calibration benchmark and validates numerical solvers, but imposes functional form restrictions violated by the piecewise-linear VRR curve in the PJM application. Niu et al. (2005) extend the numerical framework to generators holding fixed forward contracts, and Rudkevich (2004) develop a piecewise-linear approximation scheme that handles non-differentiabilities at capacity constraints. In this paper we follow the backward-integration approach, numerically solving the symmetric ODE using the `deSolve` package in R with the Holmberg boundary condition as the starting point.

The empirical performance of SFE simulations is evaluated by Sioshansi and Oren (2007), who compare simulated equilibria to observed prices in the ERCOT balancing market. They find that SFE tracks observed prices well at high demand but overpredicts markups at low demand, consistent with theory: strategic withholding is most profitable when residual demand is inelastic. This asymmetry also appears in our results, where the at-cap outcomes under the new VRR design—when the entire VRR curve is above the floor and residual demand is effectively inelastic—correspond to the most extreme predicted markups.

## 2.3 Market Power in Electricity Markets

The literature on market power in restructured electricity markets has developed two complementary methodologies: counterfactual simulation and structural estimation. Borenstein et al. (2002) establish the counterfactual simulation template by decomposing the price-cost gap in California's restructured wholesale market into market power and scarcity components. Using a competitive benchmark derived from reported marginal costs, they find that market power accounts for a substantial share of the observed price-cost wedge, particularly during high-demand hours when residual demand is inelastic. Their decomposition method — simulate competitive prices under the observed cost and demand conditions, then attribute the gap to strategic behavior — is the direct antecedent to the approach used in this paper for capacity auctions. Wolak (2003) extends this tradition by developing a unilateral market

power measure derived directly from electricity bid data, exploiting the observation that a pivotal supplier's residual demand curve is effectively inelastic at the market equilibrium. The pivotal supplier concept underlying Wolak's measure is the same logic codified in PJM's Residual Supply Index and Three Pivotal Supplier test.

Wolfram (1999) provides the first empirical test of SFE predictions, using detailed bid data from the British electricity spot market duopoly (National Power and PowerGen) to compare observed offers against the Nash equilibrium predictions of the Klemperer-Meyer model. She finds that actual markups fall between the Cournot and competitive benchmarks, broadly consistent with the range of SFE equilibria, but closer to Cournot than the symmetric Nash equilibrium would predict. Sweeting (2007) revisits the England and Wales market over a longer horizon (1995–2000) using a structural model and finds that offer behavior converged toward SFE predictions as generators accumulated experience with the market mechanism. Together, Wolfram and Sweeting establish that SFE provides empirically useful predictions for concentrated electricity spot markets, motivating its application here to the forward capacity auction setting.

Bushnell et al. (2008) extend the simulation approach to multiple US electricity markets and introduce vertical integration — between generation and retail supply — as a structural determinant of effective market concentration. Vertically integrated firms sell power to their own retail customers at regulated rates, reducing the incentive to withhold generation and thereby attenuating spot-market market power. Their cross-market evidence establishes that the structural features of seller positions (portfolio composition, contracting, vertical ownership) govern markup magnitudes as much as the number of firms. Wolak (2007) provides complementary evidence from the Australian NEM that forward contracts directly discipline spot bids, consistent with the Allaz-Vila mechanism discussed in Section 2.1. The capacity commitment in PJM's BRA creates a form of forward obligation that in principle disciplines the subsequent capacity-auction markup — though our results suggest this discipline is incomplete when the committed positions are themselves held by the same pivotal suppliers.

## 2.4 Residual Demand and the Baker–Bresnahan Framework

Baker and Bresnahan (1988) develop the foundational econometric framework for estimating the residual demand curve facing a single firm. The key insight is that a firm's market power is fully characterized by the elasticity of the residual demand it faces: the Lerner index equals the negative inverse of that elasticity at the equilibrium quantity. Estimating the residual demand curve — total demand minus the supply forthcoming from rival firms at each price level — requires instruments that shift rivals' costs without directly affecting the firm's own cost or demand. Baker and Bresnahan propose cost shifters for rivals as instruments and apply the framework to the US brewing industry.

In the capacity auction context, the residual demand curve facing a strategic seller is a shifted version of PJM's VRR schedule: total VRR demand minus the aggregate supply of fringe and rival-strategic sellers at each price. The critical distinguishing feature of the PJM application is that the residual demand schedule is set *administratively* by the regulator and published before each auction; sellers observe the VRR parameters directly. The Baker-Bresnahan problem of recovering the residual demand from market data does not arise here — the demand curve is known in closed form. This observation is what enables the SFE calibration strategy: given the observed VRR schedule and the number of strategic sellers identified by the $\text{RSI}_3$ test, the equilibrium supply function and implied Lerner index can be computed without instruments or demand estimation.

## 2.5 Capacity Market Design

The economics literature on capacity market design originates from a fundamental coordination problem: competitive energy markets, subject to regulatory price caps, tend to suppress the scarcity rents that would otherwise finance investment in peaking generation. Joskow and Tirole (2007) formalize this "missing money" problem and demonstrate that even under competitive conditions, energy-only markets underinvest in reliability when price caps bind during scarcity events. Joskow (2008) examines the remedies, arguing that capacity payments

can correct the underinvestment provided they are designed to replicate missing scarcity rents without creating perverse entry or exit incentives.

A central design question is whether the procurement target should be fixed at the reliability requirement—a vertical demand schedule—or downward-sloping, as in PJM's VRR mechanism. Cramton and Stoft (2005) make the case for a sloped administrative demand curve on the grounds that a vertical schedule generates extreme price volatility near the reliability threshold: small fluctuations in supply produce large swings between the price cap and zero. A sloped demand curve trades some quantity variation for price stability and provides smoother investment incentives across the capacity cycle. Hobbs et al. (2007) provides a formal dynamic analysis of the PJM Reliability Pricing Model and confirm that the VRR curve substantially reduces price volatility relative to vertical demand, at the cost of modest deviations from the target reliability level.

The uniform-price format of the BRA has also received attention. Cramton and Stoft (2007) defend the uniform-price auction over pay-as-bid alternatives, arguing that under the former, sellers have weaker incentives to shade their bids strategically and equilibrium prices are closer to the competitive outcome. Cramton and Ockenfels (2012) extend this analysis to the broader landscape of capacity market design and examine the interaction between auction format and market power. These papers identify a tension at the heart of capacity markets: the sloped administrative demand curve that dampens boom-bust price cycles is precisely the curve whose shape governs the rent-extraction opportunities available to a strategic seller.

The market power dimension of capacity auctions is examined in detail by Bowring (2013), who provides an overview of PJM's RPM with attention to the role of the Independent Market Monitor in detecting and mitigating anticompetitive behavior. Bowring documents the application of the TPS test and ACR mitigation across delivery years and LDAs, and notes that market power concerns are particularly acute in constrained LDAs where import capacity is limited. Vossler et al. (2009) use laboratory experiments to study behavioral responses to soft price caps in uniform-price auction environments and find that caps do

bind strategically—sellers' offers cluster near the cap, consistent with a focal equilibrium at the ceiling. Taken together, this literature motivates a structural approach to measuring the strategic markup that the binary TPS test can only detect but not quantify.

## 2.6 Price Controls in Oligopolistic Markets

The economics of price controls in oligopolistic markets provides the theoretical foundation for the allocative analysis in Section 7. The foundational reference is Weitzman (1974), who shows that the choice between price instruments and quantity instruments for regulatory intervention depends on the relative slopes of the marginal benefit and marginal cost schedules. When the marginal cost schedule is steep (supply is inelastic in the short run) and the marginal benefit schedule is relatively flat (demand is elastic), price instruments dominate quantity instruments because they avoid the deadweight loss from misallocating constrained supply. Capacity markets present precisely this structure: short-run supply is inelastic (capacity is fixed within the delivery year) and the administrative demand curve has a downward-sloping segment whose elasticity the regulator can calibrate. Weitzman's analysis implies that modifying the VRR price ceiling is an efficient regulatory instrument if and only if the unconstrained equilibrium price is genuinely above the socially optimal level — a question our SFE model addresses directly by providing a structural estimate of $p^*$.

Glaeser and Luttmer (2003) demonstrate that binding price controls in markets with imperfect competition generate misallocation that is qualitatively different from the standard deadweight loss triangle. When quantity must be rationed below the unconstrained equilibrium, goods are allocated to those with the strongest relationships to sellers rather than to those with the highest willingness to pay, creating efficiency losses that are a multiple of the simple triangle. In capacity markets the analogue is resource allocation across time: a binding cap suppresses the scarcity signal that should direct investment toward future delivery years with highest expected returns, creating an efficiency loss that accumulates dynamically rather than appearing as a static deadweight triangle.

In the SFE literature, Anderson and Xu (2005) extend the Klemperer-Meyer framework to incorporate both price caps and forward contracts, establishing conditions under which equilibria with caps exist and characterizing how caps modify the ODE boundary condition. Their analysis shows that a price cap does not simply truncate the equilibrium supply function: it changes the strategic problem by altering the payoff at the capacity boundary. When the cap is below the unconstrained Holmberg boundary price, firms face a modified optimization at the cap level, and the equilibrium supply function differs from the uncapped case throughout the price range, not just at the ceiling. This result is directly relevant to the PJM application: the settlement does not merely cap revenues at \$325/MW-day, it modifies the entire strategic incentive structure facing the $K = 3$ sellers.

Vossler et al. (2009) provide experimental evidence that price caps induce focal behavior in uniform-price auction environments: offers cluster near the cap even when the unconstrained equilibrium would clear below it. Applied to PJM, this suggests that the settlement floor at \$175/MW-day may itself become a focal point in future auctions with ample supply, as sellers coordinate on offering just above the floor rather than competing down to marginal cost. Finally, Cramton and Stoft (2005) emphasize that the missing money problem — the chronic revenue shortfall that motivates capacity markets in the first place — is recreated if capacity market prices are suppressed below the investment threshold. The settlement floor at \$175/MW-day represents a partial recognition of this risk: PJM characterizes the floor as "supporting near-term investment," an implicit acknowledgment that the cap side of the settlement poses the very problem that the capacity market was designed to solve.

# 3 Institutional Background

## 3.1 PJM's Reliability Pricing Model

PJM Interconnection operates the largest competitive wholesale electricity market in North America, coordinating the dispatch of more than 180,000 megawatts of generating capacity

across thirteen states and the District of Columbia. Unlike energy markets, which settle the hourly exchange of electricity, capacity markets procure commitments from generators to be available during high-demand periods in a future delivery year. These commitments address the "missing money" problem: energy market price caps suppress the scarcity rents that would otherwise finance investment in peaking generation, creating a structural shortfall in incentives to build new capacity (Joskow and Tirole, 2007; Joskow, 2008).

PJM implemented the Reliability Pricing Model (RPM) in 2007 to provide a forward capacity market with explicit locational pricing (Bowring, 2013). The primary procurement mechanism is the Base Residual Auction (BRA), in which suppliers submit capacity offers and the market clears against a downward-sloping administrative demand schedule—the Variable Resource Requirement (VRR) curve—at a uniform clearing price.[1] PJM also conducts Incremental Auctions in the months preceding each delivery year to accommodate late capacity additions and withdrawals, but these represent only a small fraction of total procurement.

Capacity procurement occurs separately for Locational Deliverability Areas (LDAs), geographic sub-regions of the PJM footprint constrained by transmission import limits. Each LDA has its own VRR curve parameterized by the LDA's local Cost of New Entry (Net CONE) and reliability requirement, and clears at a price at or above the RTO-wide clearing price. The transmission capacity available to import from outside an LDA is governed by the Capacity Emergency Transfer Limit (CETL). Our panel covers the RTO and sixteen sub-LDAs across delivery years 2021/22 through 2027/28, producing 110 LDA–delivery-year observations with observed clearing prices.

[1] Capacity Performance reforms introduced in 2015–2016 created a two-product structure (Base Capacity and Capacity Performance), but the BRA mechanism and the VRR curve remain central to price formation throughout our sample.

## 3.2 The Variable Resource Requirement Curve

The VRR curve is the administrative demand schedule that PJM presents to sellers in each BRA. Its shape is determined by two parameters: the Net Cost of New Entry (Net CONE) and the reliability requirement. Net CONE represents the annualized cost of building and operating a reference generating unit, net of projected revenues from energy and ancillary service markets. It is computed independently for each LDA and updated each delivery year based on cost surveys and projected market revenues.

Under the old VRR design in place for delivery years 2021/22 through 2025/26, the schedule was defined by three anchor points.[2] Point (a), the price cap, was set at $1.5 \times$ Net CONE at a quantity slightly below the reliability requirement. Point (b) was set at $0.75 \times$ Net CONE at the reliability requirement itself. Point (c) set the floor price to zero at a quantity slightly above the reliability requirement. Between anchor points the schedule is interpolated linearly, producing a steeply downward-sloping segment from Point (a) to Point (b) and a shallower segment from Point (b) to the zero-price floor.

The economic rationale for a downward-sloping demand schedule, rather than a vertical demand set at the reliability requirement, is to moderate the extreme price volatility that vertical demand generates when supply is inelastic near the reliability threshold (Cramton and Stoft, 2005; Hobbs et al., 2007). A sloped demand schedule trades some quantity uncertainty for price stability: the clearing price falls smoothly as surplus capacity rises, rather than collapsing discontinuously from the cap to zero once the reliability requirement is met. This design was specifically advocated as a capacity market reform to improve investment incentives and dampen boom-bust price cycles (Cramton and Ockenfels, 2012).

Because the VRR parameters are publicly announced well in advance of the auction, the demand side of the market is fully transparent to sellers. Every participant observes the price-quantity pairs of all anchor points before submitting offers. This transparency is a

[2] We distinguish the "old" design from the "new" design introduced by FERC Order ER25-1357 for delivery years 2026/27 and beyond; see Section 3.3.

defining feature of the PJM capacity market and distinguishes it from environments with demand uncertainty—a distinction that proves consequential for equilibrium selection in the SFE framework developed in Section 4.

### 3.3 The 2026 VRR Redesign

FERC approved a restructured VRR curve design in 2025 (Order ER25-1357), taking effect with the 2026/27 delivery year BRA held in July 2025. The redesign introduced a fourth anchor point that creates an explicit price floor. Under the new design:

- Point (a) remains the price cap, set at approximately $1.5 \times$ Net CONE;
- A downward-sloping segment connects Point (a) to Point (b), the floor price (approximately \$177–\$179 per MW-day in our sample);
- A flat segment at the floor price extends from Point (b) to Point (c), so that additional capacity clears at the floor rather than at zero;
- Point (d) marks the end of the flat floor segment at a quantity well above the reliability requirement.

The key economic change is the elimination of the collapse to a zero price floor. By setting a hard lower bound of approximately \$177–\$179 per MW-day—close to the projected net energy and ancillary service revenue available to a new entrant—the redesign prevents prices from falling below the level at which new investment would break even, even in conditions of surplus capacity.

Table 1 documents the BRA timeline and RTO clearing outcomes for all seven completed auctions in our sample. Two facts stand out. First, both auctions conducted under the new VRR design—the 2026/27 BRA (July 2025) and the 2027/28 BRA (January 2026)—cleared at exactly the Point (a) price cap: \$329.17 and \$333.44 per MW-day, respectively. This "at-cap" outcome, in which aggregate supply intersects the VRR at its uppermost anchor point, is

consistent with strategic equilibrium pricing and motivates the supply function equilibrium analysis in Sections 4–6. Second, the RTO clearing price rose sharply between 2024/25 ($28.92 per MW-day) and 2025/26 ($269.92), a price spike that precedes the VRR redesign and suggests a structural shift in capacity market conditions beginning with the 2025/26 delivery year. The contrast between the old-design auctions—which exhibited outcomes ranging from $28.92 to $269.92 per MW-day—and the new-design auctions, which uniformly cleared at the cap, provides a natural experiment for the comparative static in Section 6.3.

**Table 1:** PJM Base Residual Auction Timeline and RTO Clearing Outcomes, 2021/22–2027/28

| Delivery Year | BRA Date | Lead (mo.) | Clearing ($/MW-day) | Net CONE ($/MW-day) | VRR Design | At Cap |
|---|---|---|---|---|---|---|
| 2021/22 | May 2018 | 37.0 | $140.00 | $321.57 | Old | No |
| 2022/23 | May 2021 | 12.5 | $50.00 | $260.50 | Old | No |
| 2023/24 | July 2022 | 11.0 | $34.13 | $274.96 | Old | No |
| 2024/25 | May 2024 | 0.8 | $28.92 | $293.19 | Old | No |
| 2025/26 | August 2024 | 9.9 | $269.92 | $228.81 | Old | No |
| 2026/27 | July 2025 | 10.2 | $329.17 | $212.14 | New | **Yes** |
| 2027/28 | January 2026 | 16.8 | $333.44 | $242.52 | New | **Yes** |

*Note:* Lead time is the number of calendar months between the BRA settlement date and the start of the delivery year. RTO clearing prices and Net CONE values are in nominal dollars per MW-day. "At Cap" indicates that the RTO clearing price equaled the Point (a) price cap under the relevant VRR design. Data sources: PJM Planning Parameters; PJM BRA Results.

## 3.4 Auction Lead-Time Irregularity

A second distinctive feature of the PJM capacity market during this period is the extreme irregularity of the BRA schedule. In a well-functioning capacity market, auctions should be held well in advance of the delivery year—typically two to three years ahead—to provide investment signals with enough lead time for new generation to be financed, permitted, and

constructed (Hobbs et al., 2007). The PJM RPM was designed with a three-year forward horizon.

Table 1 reveals substantial deviations from this design. The 2021/22 BRA was held in May 2018, 37 months before the start of delivery—consistent with the intended horizon. However, a series of FERC-mandated market reforms delayed subsequent auctions, compressing lead times to 12.5 months for 2022/23 and 11 months for 2023/24. No BRA was held at all in calendar year 2023. Most strikingly, the 2024/25 BRA was conducted in May 2024, a mere 0.8 months before the June 2024 start of delivery— effectively a spot capacity auction. Under such conditions, no new generation could plausibly have been built in response to the price signal; the auction served purely as a spot-market procurement exercise rather than an investment coordination mechanism.

The disrupted schedule has two implications for our analysis. First, the 2024/25 delivery year should not be interpreted as a period in which the capacity market was providing forward investment incentives, and the low clearing price of $28.92 per MW-day that year reflects the absence of a forward capacity shortage rather than an equilibrium outcome of a properly functioning forward market. Second, the compressed lead times for 2022/23 through 2024/25 may have contributed to the sharp price increase observed in 2025/26: if new entrants were unable to respond to scarcity signals in time for those delivery years, the accumulated capacity shortfall would manifest as high prices in subsequent auctions. Our SFE simulation takes the VRR curve and capacity endowments as given and does not model the investment response to price signals, but this context is important when interpreting the 2024/25 and 2025/26 results. The relationship between lead-time irregularity and the price-signaling function of capacity auctions is examined in detail in Section 8.

### 3.5 Market Power Mitigation: The Three Pivotal Supplier Test

PJM's primary market power mitigation mechanism in capacity auctions is the Three Pivotal Supplier (TPS) test. Supplier $i$ is defined as *pivotal* in LDA $\ell$ if removing $i$'s capacity from

the market would cause aggregate supply to fall short of the reliability requirement— that is, if the market cannot clear without $i$'s participation. The TPS test asks whether any individual seller, or any coalition of the three largest sellers, is pivotal. When the test is triggered, the affected suppliers' offers are capped at the Avoidable Cost Rate (ACR), a measure of short-run variable cost (Bowring, 2013).

The Residual Supply Index for $k$ suppliers in LDA $\ell$ is defined as

$$\mathrm{RSI}_k^\ell = \frac{\text{Total capacity in LDA } \ell \; - \; \text{Capacity of top-}k \text{ sellers in LDA } \ell}{\text{Reliability requirement for LDA } \ell}. \tag{1}$$

An RSI below unity means the market outside the top-$k$ sellers cannot meet the reliability requirement on its own, confirming that those sellers are collectively pivotal. Mitigation is triggered whenever $\mathrm{RSI}_1 < 1$ or $\mathrm{RSI}_3 < 1$.

Table 2 summarizes TPS test outcomes for the RTO and selected LDAs across the delivery years for which PJM IMM State of the Market reports provide RSI data. Three patterns stand out. First, $\mathrm{RSI}_3$ falls below unity for every LDA and every delivery year in the panel, so the TPS mitigation threshold is met throughout our entire sample. Second, the concentration is strikingly persistent: in the RTO, $\mathrm{RSI}_3$ ranges only from 0.62 to 0.73 across six delivery years despite substantial year-to-year variation in clearing prices (\$28.92 to \$333.44 per MW-day). Third, the most extreme concentration appears in small sub-LDAs: BGE and DEOK each report a single seller with $\mathrm{RSI}_1 = 0$ throughout the sample, implying that those LDAs depend entirely on one supplier to meet their reliability requirement.

**Table 2:** TPS Test Results: Residual Supply Indices by LDA and Delivery Year

| Delivery Year | LDA | $RSI_1$ | $RSI_3$ | Participants | All Pivotal |
|---|---|---|---|---|---|
| 2022/23 | RTO | 0.81 | 0.73 | 130 | Yes |
| 2022/23 | MAAC | 0.69 | 0.37 | 25 | Yes |
| 2022/23 | EMAAC | 1.25 | 0.64 | 7 | Yes |
| 2022/23 | COMED | 0.43 | 0.36 | 14 | Yes |
| 2022/23 | BGE | 0.00 | 0.00 | 1 | Yes |
| 2022/23 | DEOK | 0.00 | 0.00 | 1 | Yes |
| 2023/24 | RTO | 0.78 | 0.68 | 134 | Yes |
| 2023/24 | MAAC | 0.78 | 0.40 | 11 | Yes |
| 2024/25 | RTO | 0.77 | 0.64 | 133 | Yes |
| 2024/25 | MAAC | 0.59 | 0.11 | 9 | Yes |
| 2024/25 | EMAAC | 0.48 | 0.00 | 2 | Yes |
| 2025/26 | RTO | 0.82 | 0.62 | 128 | Yes |
| 2026/27 | RTO | 0.82 | 0.64 | 153 | Yes |
| 2027/28 | RTO | 0.82 | 0.63 | 155 | Yes |

*Note:* $RSI_k$ is the Residual Supply Index for the top-$k$ sellers; see equation (1). "All Pivotal" indicates that every participant is individually pivotal (i.e., $RSI_1 < 1$ when computed for each seller individually). Data source: PJM IMM State of the Market reports, various years.

Critically, the TPS test is a binary instrument: it identifies whether mitigation conditions are met but says nothing about the magnitude of strategic markup or the size of the resulting price-cost margin. In the RTO, $RSI_3$ has been persistently low and essentially stable across six delivery years—yet clearing prices have ranged from \$28.92 to \$333.44 per MW-day, a factor of eleven. The binary TPS test cannot distinguish the near-competitive outcome in 2023/24 from the at-cap outcomes in 2026/27 and 2027/28. The supply function equilibrium framework developed in Section 4 provides a structural model that generates continuous,

quantitative price-cost margin estimates and can explain this price variation through changes in the VRR curve parameters and the market structure documented here.

## 3.6 The Shapiro Settlement: A Modified Demand Curve for 2026/27–2027/28

The high BRA clearing prices of 2025/26 and the anticipation of similarly elevated prices in subsequent auctions triggered a political intervention that directly modified the demand curve for two delivery years. On December 30, 2024, the Office of Governor Josh Shapiro filed a complaint against PJM's tariff with the Federal Energy Regulatory Commission under Section 206 of the Federal Power Act. The complaint's legal theory was that the VRR Point (a) price cap was set too high given lower expectations for new entry in the near term (Bresler and Keech, 2025). Because compressed auction lead times (Section 3.4) limited the ability of new generation to respond to high prices within upcoming delivery periods, the complaint argued that high prices would not produce efficient entry and instead represented an unjust and unreasonable rate. The proposed remedy was to lower the cap to $1.5 \times$ Net CONE and remove reliance on Gross CONE in the Point (a) calculation.

Following confidential settlement discussions, PJM and the Governor's office announced an agreement in principle on January 28, 2025 (Office of Governor Josh Shapiro, 2025). The settlement was presented to the PJM Markets and Operations Committee on February 7, 2025, and PJM indicated a target filing date of February 14, 2025, with the 2026/27 BRA scheduled for July 2025 and the 2027/28 BRA for December 2025 (Bresler and Keech, 2025).

**Settlement terms.** The agreement imposes a price band on the demand curves for both delivery years:

- **Price cap**: approximately \$325/MW-day UCAP (Unforced Capacity), equivalent to \$256.75/MW-day in ICAP terms at the assumed 79% accreditation of the dual-fuel combustion turbine reference unit.

- **Price floor**: approximately \$175/MW-day UCAP (\$138.25/MW-day ICAP), below which all available capacity clears regardless of the VRR slope.

We denote the settlement cap as $\bar{p}_S \approx$ \$325/MW-day (UCAP) and the floor as $\bar{p}_f \approx$ \$175/MW-day (UCAP) to distinguish them from the existing VRR Point (a) cap $\bar{p}$. Both values are indexed to the accreditation of the reference unit and may differ slightly from these round numbers when final ELCC values are determined.

**VRR curve mechanism.** The settlement does not impose an external price constraint on an otherwise unchanged auction. Instead, it modifies the shape of the VRR demand curve itself (Bresler and Keech, 2025):

1. At prices above $\bar{p}_S$, the demand curve is flat (no capacity with offers above the cap can clear);
2. Between $\bar{p}_S$ and $\bar{p}_f$, the existing VRR sloped segment applies;
3. At prices at or below $\bar{p}_f$, the demand curve extends horizontally to clear all available capacity at the floor price.

This mechanism applies identically to the RTO demand curve and to every LDA demand curve. Clearing results continue to respect CETL transmission limitations (Bresler and Keech, 2025).

**The two-cap distinction.** The settlement introduces a second layer of price constraint on top of the VRR design already in place. The pre-existing Point (a) cap $\bar{p}$ is calibrated to investment incentives (approximately 1.5 × Net CONE) and has been part of PJM's market design since 2007. The settlement cap $\bar{p}_S$ is lower and is motivated by near-term consumer cost concerns rather than investment-adequacy principles. Table 1 shows that $\bar{p} =$ \$329.17/MW-day in 2026/27 — only \$4.17 above $\bar{p}_S =$ \$325/MW-day. At the RTO level, the settlement is barely binding; the effect is largest in constrained LDAs where local

Point (a) caps substantially exceeded \$325 in prior years. The analysis in Sections 6–10 compares the SFE equilibrium price $p^*$ (computed against the unmodified VRR curve) to both caps, establishing where the settlement is and is not binding.

# 4 Model

## 4.1 Setup

The BRA is modeled as a static, complete-information game in which sellers simultaneously submit supply functions before the auction clears. This formulation follows Klemperer and Meyer (1989) and, for the electricity market application, Green and Newbery (1992). The timing is: (1) PJM publishes the VRR curve parameters for the upcoming delivery year; (2) sellers submit supply functions; (3) PJM clears the auction at the uniform price where aggregate supply meets the VRR curve; (4) sellers deliver committed capacity during the delivery year.

**Assumption 1.** *There are $K \geq 2$ strategic sellers, indexed $i = 1, \ldots, K$, and a competitive fringe. Each strategic seller $i$ holds a fixed capacity portfolio $\bar{q}_i > 0$ measured in megawatts. A supply function $S_i : [0, \bar{p}] \to [0, \bar{q}_i]$ is a nondecreasing, twice continuously differentiable mapping from price to quantity. Each seller simultaneously commits to a supply function before the auction clears.*

**Assumption 2.** *The competitive fringe bids at its avoidable cost rate $c_f > 0$. Fringe supply is $S_f(p) = 0$ for $p < c_f$ and $S_f(p) = \bar{Q}_f$ for $p \geq c_f$, where $\bar{Q}_f > 0$ is total fringe capacity. The fringe is price-taking and its behavior is common knowledge.*

**Assumption 3.** *Each strategic seller has a constant marginal cost $c > 0$ for output $q \leq \bar{q}_i$ and infinite marginal cost above capacity. For the symmetric calibration we set $c$ equal to the sector-average avoidable cost rate published by the PJM IMM.*

The market clears at the price $p^*$ satisfying

$$\sum_{i=1}^{K} S_i(p^*) + S_f(p^*) = D(p^*), \tag{2}$$

where $D(\cdot)$ is the VRR demand schedule described below. Every seller receives the same clearing price $p^*$ for its committed capacity. The *residual demand* facing seller $i$ at price $p$ is

$$R_i(p) \;=\; D(p) - \sum_{j \neq i} S_j(p) - S_f(p). \tag{3}$$

In equilibrium, $S_i(p^*) = R_i(p^*)$. Because $D(\cdot)$ and all rivals' strategies are publicly known before the auction, there is no demand uncertainty: the residual demand curve each seller faces is deterministic.[3] This determinism means firms observe the VRR parameters before submitting offers; the boundary condition is imposed as an equilibrium selection criterion, as discussed in Section 4.4.

## 4.2 The VRR Demand Schedule

PJM's Variable Resource Requirement curve is a piecewise-linear, publicly-announced downward-sloping demand schedule. Under the old design (delivery years 2021/22–2025/26) it is parameterized by three anchor points $(p_a, q_a)$, $(p_b, q_b)$, and $(0, q_c)$:

$$D(p) = \begin{cases} q_a & p \geq p_a \\ q_a + \dfrac{q_b - q_a}{p_b - p_a}(p - p_a) & p_b \leq p < p_a \\ q_b + \dfrac{q_c - q_b}{0 - p_b}(p - p_b) & 0 \leq p < p_b, \end{cases} \tag{4}$$

[3] Holmberg's 2008 uniqueness proof formally requires stochastic demand with positive probability of all capacity being demanded. In the deterministic BRA setting, the boundary condition $s(\bar{p}) = \bar{q}$ is imposed as a selection criterion motivated by that result. The at-cap clearing outcomes in 2026/27 and 2027/28 confirm the condition is active in practice.

where $p_a = 1.5 \times$ Net CONE is the price cap and $p_b = 0.75 \times$ Net CONE is the reliability-requirement price. The quantities $q_a < q_b < q_c$ are the corresponding capacity levels, with $q_b$ equal to the reliability requirement. Because $D(p) = q_a$ for all $p \geq p_a$, the VRR curve is effectively inelastic above the cap.

Under the new design (delivery years 2026/27 onward), FERC Order ER25-1357 introduces a fourth anchor point $(p_f, q_f)$ and $(p_f, q_d)$ with $q_f > q_b$, creating a flat floor segment at price $p_f \approx$ \$177– \$179/MW-day. The new schedule is:

$$D(p) = \begin{cases} q_a & p \geq p_a \\ q_a + \dfrac{q_b - q_a}{p_f - p_a}(p - p_a) & p_f \leq p < p_a \\ q_d & 0 \leq p < p_f. \end{cases} \tag{5}$$

The flat floor segment at $p_f$ means that clearing prices cannot fall below $p_f$ along the VRR, changing the elasticity profile that sellers face and thus the equilibrium markup. The comparative static in Section 6.3 isolates this design change.

The slope $D'(p)$ is piecewise constant and satisfies:

**Assumption 4.** *$D'(p) < 0$ on the sloped segments; $D'(p) = 0$ on the flat segments (at and above $p_a$, and below $p_f$ under the new design). The function $D(\cdot)$ is known to all players before the auction.*

## 4.3 Firm Optimization and the SFE Condition

Seller $i$ chooses $S_i(\cdot)$ to maximize profit. Taking rivals' supply functions as fixed, the profit at any market-clearing price $p$ is $\pi_i = p \cdot S_i(p) - c \cdot S_i(p) = (p - c) \cdot S_i(p)$, since marginal cost is constant at $c$. The seller's problem is equivalent to maximizing profit along its residual demand curve $R_i(\cdot)$. At price $p$, if seller $i$ considers increasing its offered quantity by $\mathrm{d}q$, the

market-clearing price must fall by d$p$ to preserve equation (2):

$$\mathrm{d}p \;=\; \frac{\mathrm{d}q}{D'(p) - \sum_{j\neq i} S_j'(p) - S_f'(p)}. \tag{6}$$

Since $D'(p) < 0$ and $S_j'(p) \geq 0$, the denominator is strictly negative: an increase in supply lowers the clearing price. The seller's marginal revenue from increasing quantity at price $p$ is:

$$\mathrm{MR}_i(p) \;=\; p + \frac{S_i(p)}{D'(p) - \sum_{j\neq i} S_j'(p) - S_f'(p)}. \tag{7}$$

Setting marginal revenue equal to marginal cost $c$ yields the *supply function equilibrium condition*:

**Proposition 1.** *In a supply function equilibrium, each seller $i$'s strategy $S_i(\cdot)$ satisfies*

$$S_i(p) \;=\; -\Big(p - c\Big)\Big[D'(p) - \textstyle\sum_{j\neq i} S_j'(p) - S_f'(p)\Big] \tag{8}$$

*at every price $p$ at which the market can clear. This is a system of $K$ coupled first-order ordinary differential equations in the supply functions $\{S_1, \ldots, S_K\}$.*

Equation (8) is the $K$-seller generalization of the first-order condition derived by Green and Newbery (1992) for the symmetric duopoly (their equation 3) and of the general condition in Anderson and Hu (2008) (their equation 1). Its economic content is straightforward: seller $i$ equates its residual-demand markup $(p - c)$ to the inverse of the residual demand elasticity at that price. When more sellers compete ($K$ large) or the VRR slope is steep (large $|D'|$), the system requires a smaller markup at each price, pushing the equilibrium supply function toward the competitive schedule.

## 4.4 Equilibrium Selection and the Boundary Condition

The ODE system (8) generally admits a continuum of solutions. Klemperer and Meyer (1989) show that under demand uncertainty, the set of symmetric supply function equilibria is an

interval bounded above by the Cournot schedule (where each firm behaves as if rivals are at capacity) and below by the competitive schedule (where $p = c$). Any trajectory between these two stationary solutions is a Nash equilibrium. This multiplicity complicates both prediction and comparative statics.

The multiplicity is resolved by the following result, which applies directly to the capacity-auction setting:

**Proposition 2** (Holmberg 2008)**.** *Suppose sellers are symmetric, demand is inelastic above the price cap $\bar{p}$, and the probability that aggregate demand reaches the capacity ceiling is strictly positive. Then there exists a unique symmetric supply function equilibrium. In this equilibrium each seller's supply function satisfies the boundary condition*

$$S_i(\bar{p}) \;=\; \bar{q}_i \quad \textit{for all } i. \tag{9}$$

The intuition is direct. When rivals' capacity constraints simultaneously bind, seller $i$ faces perfectly inelastic residual demand. Its monopoly optimum at that moment is to set price equal to the cap $\bar{p}$; any lower price sacrifices revenue without reducing quantity (since residual demand is inelastic), and any higher price is infeasible. This forces the supply function to pass through the point $(\bar{p}, \bar{q}_i)$, tying the arbitrary constant in the ODE and selecting a unique trajectory.

In the PJM BRA, the price cap $\bar{p}$ equals the VRR Point (a) price, and Assumption 4 ensures $D'(\bar{p}) = 0$ (the VRR is flat at the cap). The at-cap clearing outcomes in 2026/27 and 2027/28 (Table 1) confirm that the boundary condition is active in those years, consistent with Holmberg's (2008) selection criterion: PJM's reliability target requires procuring essentially all installed capacity, making binding-capacity outcomes the empirically relevant case. We therefore integrate the ODE (8) *backward* from the boundary condition $S_i(\bar{p}) = \bar{q}_i$, following the solution method of Green and Newbery (1992).

## 4.5 The Symmetric Special Case

For calibration and comparative statics we begin with the symmetric case in which all $K$ strategic sellers are identical: $\bar{q}_i = \bar{q}$ and $c_i = c$ for all $i$. In a symmetric Nash equilibrium, $S_i(p) = s(p)$ for all $i$. Substituting into equation (8):

$$s(p) \;=\; -\Big(p - c\Big)\Big[D'(p) - (K-1)\,s'(p) - S_f'(p)\Big]. \tag{10}$$

Under Assumption 2, $S_f'(p) = 0$ almost everywhere. Rearranging (10) for $s'(p)$ yields the *symmetric SFE ordinary differential equation*:

$$\boxed{s'(p) \;=\; \frac{1}{K-1}\left[D'(p) + \frac{s(p)}{p-c}\right], \quad p \in [c,\, \bar{p}),} \tag{11}$$

with boundary condition $s(\bar{p}) = \bar{q}$. This is a first-order, nonautonomous ODE in the single unknown function $s(\cdot)$. It is the $K$-firm generalization of Green and Newbery's equation (3), to which it reduces when $K = 2$ and $S_f \equiv 0$.

Equation (11) has a direct economic interpretation. The term $D'(p)$ captures the direct effect of price on quantity demanded: a steeper VRR slope (larger $|D'|$) makes $s'(p)$ more negative, meaning each firm's equilibrium supply function falls more steeply toward the cap as price rises. The term $s(p)/(p-c)$ is a markup-boundary feedback: as $p$ approaches $c$ from above, this term diverges and forces $s'(p)$ to remain bounded away from the demand slope, enforcing the lower boundary of the supply function and preventing equilibrium offers below cost. At prices well above $c$, the feedback term partially offsets $D'(p)$, making the supply function steeper in price than a linear interpolation between $c$ and $\bar{p}$ would predict. More competitors (larger $K$) divide the denominator, flattening each firm's supply function toward the competitive benchmark.

The equilibrium clearing price $p^*$ is found by solving

$$K \cdot s(p^*) + S_f(p^*) = D(p^*) \tag{12}$$

for $p^*$ after integrating (11) backward from $\bar{p}$. The implied price-cost margin (Lerner index) is

$$L = \frac{p^* - c}{p^*}, \tag{13}$$

which serves as the primary measure of strategic markup (price-cost margin) in the results of Section 6. The competitive benchmark corresponds to $c = p^*$ ($L = 0$), attained as $K \to \infty$ or as $|D'(p)| \to \infty$ (perfectly elastic demand). The Cournot upper bound on the markup is $L^{\text{Cournot}} = 1/(K \cdot |D'(p^*) \cdot p^*/D(p^*)|)$, the standard inverse-elasticity rule evaluated at market demand (which equals the residual demand elasticity in the symmetric case).

**The symmetry assumption.** In reality, the three largest PJM capacity sellers hold heterogeneous portfolios: a mix of baseload nuclear, peaking gas, renewables, and demand response with widely differing cost structures and capacity sizes. Treating them as symmetric is a deliberate first-order approximation, not a statement about the actual market. The justification is tractability: the symmetric case reduces the $K$-dimensional system of coupled ODEs in equation (8) to the single scalar ODE (11), making the model calibratable from publicly available aggregate data. This is the standard approach in the calibrated SFE literature (Green and Newbery, 1992; Sioshansi and Oren, 2007). The direction of the asymmetry bias on the average price-cost margin is ambiguous: a lower-cost seller has a higher markup than the symmetric benchmark; a higher-cost seller has a lower one. On average, these effects partially offset. We treat the symmetric Lerner as an estimate of the capacity-weighted average markup across the strategic sellers, and note that the spread around this average is an object of interest for future asymmetric extensions.

Equation (11) is solved numerically using a fourth-order Runge–Kutta (RK4) integrator with fixed step size $h = 0.05\,\$/\text{MW-day}$. To improve stability we apply the substitution $\tau = \bar{p} - p$, which converts the backward initial-value problem into a standard forward IVP with $s(0) = \bar{q}$. The piecewise nature of $D'(p)$ is accommodated by restarting the integration at each VRR kink point: the integration grid is split at $p_b$ (old design) or $p_f$ (new design),

so the integrator never crosses a discontinuity in $D'(p)$ within a single step. Under the new design, $D(p)$ has a jump discontinuity at $p_f$ (from $q_b$ approaching from above to $q_d > q_b$ below); equilibrium is resolved by identifying $p^*$ as the price at which aggregate supply meets the sloped-segment demand from above, which coincides with $p_f$ when $K \cdot s(p_f) + S_f(p_f) = q_b$. Full details of the numerical implementation are provided in Section 5.

# 5 Calibration

## 5.1 Data Sources

The calibration draws on four publicly available datasets produced by PJM and the PJM Independent Market Monitor (IMM).

**PJM Planning Parameters.** PJM publishes the planning parameters for each BRA in advance of the auction, including the VRR curve anchor points, Net CONE, CETL, and reliability requirement for every LDA. These files are available in Excel format and provide the complete VRR demand schedule that sellers observe when submitting capacity offers. We processed the planning parameter files for delivery years 2021/22 through 2028/29 using a custom parser that extracts LDA-level parameters for each VRR design vintage and handles the three-point (old) and four-point (new) curve specifications separately. Spot-checks against published PJM summary tables confirm accuracy (e.g., 2023/24 RTO Net CONE = \$274.96 per MW-day).

**PJM BRA Results.** Following each auction, PJM releases clearing prices and quantities by LDA. We processed seven completed auctions (2021/22 through 2027/28); the 2028/29 BRA had not yet been conducted at the time of data collection. Clearing results are verified against PJM press releases (e.g., 2025/26 RTO clearing = \$269.92 per MW-day; 2026/27 RTO clearing = \$329.17).

**PJM IMM State of the Market Reports.** The PJM IMM publishes an annual State of the Market (SotM) report containing auction-by-auction summaries of market concentration, TPS test outcomes, and offer-curve characteristics. We extracted Residual Supply Index values and participant counts from the relevant report volumes using a PDF-parsing pipeline. Volume 2022 covers delivery years through 2022/23, and Volume 2025 covers delivery years 2023/24 through 2027/28. Due to layout variation across report vintages, RSI data is available for a subset of LDAs and delivery years, as documented in Table 2. The 2021/22 delivery year predates our RSI extraction window and is excluded from the market structure analysis.

**Auction Lead Times.** BRA settlement dates and delivery-year start dates were coded manually from PJM auction notices and verified against the data in Table 1. Lead time is defined as the number of calendar months between the BRA settlement date and the first day of the delivery year. The highly irregular lead-time pattern documented in Section 3.4 is incorporated as a covariate in the calibration of the investment-signal environment.

The four datasets are joined at the LDA–delivery-year level to produce a master panel with 110 observations spanning 17 LDAs and seven delivery years (2021/22 through 2027/28).

## 5.2 VRR Curve Parameters

Table 3 reports the VRR curve parameters for the RTO under both the old and new designs for the six delivery years with complete BRA data. Under the old design, Point (a) equals $1.5\times$Net CONE and Point (b) equals $0.75\times$Net CONE by construction, so the ratio $p_a/p_b = 2$ is exact for all old-design years except 2025/26, where a transitional administrative cap produced a Point (a) price above the standard $1.5 \times$ Net CONE multiple.[4] Under the new design, Points (b) and (c) equal the explicit floor price of approximately \$177–\$179 per

[4]The 2025/26 Net CONE of \$228.81 per MW-day implies a standard cap of \$343.22, but the published Point (a) was \$451.61. This discrepancy reflects the fact that VRR anchor points are fixed at planning-parameter publication and may diverge from the final Net CONE if energy and ancillary service revenues are revised substantially between publication and auction. We use the published anchor points as the realized VRR parameters in all calibrations.

MW-day, and Point (a) is calibrated by FERC Order ER25-1357.[5]

**Table 3:** RTO VRR Curve Parameters, 2022/23–2027/28

| Delivery Year | Net CONE ($/MW-day) | Pt. (a) Price | Pt. (a) MW | Pt. (b) Price | Pt. (b) MW | VRR Design |
|---|---|---|---|---|---|---|
| 2022/23 | $260.50 | $390.75 | 136,076 | $195.38 | 139,656 | Old |
| 2023/24 | $274.96 | $412.44 | 135,914 | $206.22 | 139,473 | Old |
| 2024/25 | $293.19 | $439.79 | 138,341 | $219.89 | 141,910 | Old |
| 2025/26 | $228.81 | $451.61 | 133,554 | $171.61 | 137,160 | Old |
| 2026/27 | $212.14 | $329.17 | 134,887 | $177.24 | 136,361 | New |
| 2027/28 | $242.52 | $333.44 | 141,681 | $179.55 | 143,218 | New |

*Note:* Prices in nominal dollars per MW-day; quantities in MW. Under the old design, Point (c) is the zero-price floor at a quantity above Point (b); under the new design, Points (b) and (c) share the floor price and a flat segment extends to Point (d). The 2025/26 Point (a) price exceeds 1.5 × Net CONE due to a transitional administrative cap (see footnote above). Data source: PJM Planning Parameters.

The panel encompasses 110 LDA–delivery-year cells with observed clearing prices across 17 LDAs. The three benchmark calibration years used throughout Sections 4 and 6 are selected to represent three distinct capacity market regimes: *2023/24*, a low-price year under the old VRR design exhibiting equilibrium prices consistent with competitive pricing under TPS mitigation ($34.13 per MW-day, $\text{RSI}_3 = 0.68$); *2025/26*, a high-price year under the old VRR design that preceded the redesign ($269.92 per MW-day, $\text{RSI}_3 = 0.62$); and *2026/27*, the first auction under the new VRR design, which cleared at exactly the Point (a) cap

[5]The new-design Point (a) prices ($329.17 for 2026/27, $333.44 for 2027/28) are not exactly 1.5×Net CONE (1.5 × $212.14 = $318.21 and 1.5 × $242.52 = $363.78 respectively). The $329.17 figure reflects the settlement implementation: FERC Order ER25-1357, which implemented both the new VRR design and the settlement's cap provision, produced a Point (a) approximately $4 above the nominal settlement cap of $325/MW-day UCAP. The precise derivation follows the same dynamic as the 2025/26 deviation (see footnote above): VRR anchor points are fixed from planning parameters published before the auction and may incorporate accreditation adjustments that produce a Point (a) differing from a simple Net CONE multiple. We treat the published $329.17 and $333.44 as the effective VRR boundary conditions and note that the settlement's flat-top at $325 creates a binding gap of $4.17 and $8.44 respectively below these Point (a) values.

($329.17 per MW-day, $\mathrm{RSI}_3 = 0.64$). Together, these three years span the full range of price outcomes in the sample and allow the comparative static analysis to isolate the effect of VRR design changes from variation in market structure and cost parameters.

## 5.3 Market Structure

Capacity market concentration in PJM is high by any standard measure. The PJM IMM reports LDA-level Herfindahl-Hirschman Index (HHI) values in its annual State of the Market (SotM) reports; in every delivery year covered by our sample, the RTO-level HHI exceeds 2,500 and most constrained LDAs exceed 4,000, thresholds the Department of Justice classifies as highly concentrated markets under standard antitrust benchmarks.

The key calibration input is the number of strategic sellers, $K$. We use the Residual Supply Index with threshold three ($\mathrm{RSI}_3$) as the calibration criterion: $\mathrm{RSI}_3 < 1$ means that the three largest sellers collectively hold pivotal capacity, i.e., no competitive outcome is supportable without their participation. In all three benchmark years, the RTO-level $\mathrm{RSI}_3$ falls below unity (0.68, 0.62, and 0.64 for 2023/24, 2025/26, and 2026/27, respectively), establishing that at most three large sellers are jointly pivotal. The 2026/27 at-cap clearing outcome further rules out $K \geq 4$: our model predicts interior clearing for four or more strategic sellers in that year (Section 6.2), so the observed cap clearing is only consistent with $K \leq 3$. The IMM TPS test confirms individual pivotality for all three largest sellers in every constrained LDA, supporting $K = 3$ over $K = 2$ as the baseline. We therefore set $K = 3$ as the baseline strategic count, and examine sensitivity to this assumption via the $K$ comparative static in Section 6.2. Because RSI data is available only at the RTO level for most delivery years, we impose the same $K = 3$ structure on each LDA calibration and allow cross-LDA variation to enter through the VRR parameters and CETL-based fringe capacity.[6]

[6] LDA-level TPS test outcomes reported in the IMM SotM confirm that at least three suppliers pass the pivotal test in all constrained LDAs, consistent with $K = 3$.

## 5.4 Cost Parameters

The avoidable cost rate (ACR) for each strategic seller is set to $c = \$150$ per MW-day, the combined-cycle sector average reported in the 2025 PJM IMM State of the Market (Table 3-8). Combined-cycle units represent the marginal capacity technology in PJM's current resource mix, and the IMM identifies the combined-cycle ACR as the relevant benchmark for capacity-offer mitigation under the TPS test. The symmetric-firm assumption sets all $K = 3$ sellers to the same $c$; heterogeneity in costs and capacity sizes is abstracted away. It is important to note that the ACR is a proxy for opportunity cost, not its direct measure. The true avoidable cost of offering a megawatt of capacity depends on projected energy-market revenues, fuel hedge positions, maintenance schedules, and the firm's financial structure — none of which are publicly observable at the portfolio level. The sector-average combined-cycle ACR therefore understates cost heterogeneity across strategic sellers. The sensitivity of equilibrium markups to this approximation is examined in Section 6.4, where fringe supply is scaled by a factor $\gamma \in [0.5, 2.0]$ around the baseline; this implicitly tests robustness to a wide range of effective cost assumptions.

Fringe supply is modeled as an inelastic block offered at $c$: any capacity not controlled by the $K$ strategic sellers is assumed to offer at cost. At the RTO level, fringe capacity is set to total installed capacity minus the capacity attributed to the three largest sellers; this residual is scaled with $\gamma$ to explore fringe-size sensitivity. For individual LDA calibrations, the Capacity Emergency Transfer Limit (CETL) serves as the effective fringe capacity, since CETL measures the maximum deliverable import from the rest of PJM into the LDA. LDA-level CETL values by delivery year are drawn directly from the PJM planning parameter files described in Section 5.1.

## 5.5 Benchmark Delivery Years

We select three delivery years as calibration benchmarks, each representing a distinct capacity market regime. *2023/24* is a low-price interior year ($34.13/MW-day) under the old VRR

design in which PJM's TPS mitigation was broadly applied; the gap between the model's unmitigated SFE price and the observed clearing price establishes the baseline interior equilibrium for the model and illustrates the effect of offer-price mitigation. *2025/26* is a high-price year ($269.92/MW-day) that preceded the VRR redesign; it exhibits a large upward price spike relative to prior years and partial TPS application, making it useful for isolating the effect of VRR slope on equilibrium markups. *2026/27* is the first auction under the new VRR design and cleared exactly at the Point (a) cap ($329.17/MW-day), providing the structural evidence used in Section 6.1 to establish $K \leq 3$ as consistent with the observed at-cap outcome.

The three benchmark years also span the range of outcomes relative to the Shapiro settlement cap of $325/MW-day (Section 3). In 2023/24, the settlement would be entirely non-binding ($p^* = \$330$ exceeds $325 but clearing occurred at $34 under mitigation; the settlement cap is irrelevant). In 2026/27, the settlement cap is binding at the RTO level ($p^* = \$329 > \$325$). This span allows the policy analysis in Sections 7–10 to evaluate both binding and non-binding settlement scenarios.

We exclude *2024/25* from the benchmark set. PJM conducted the 2024/25 BRA as a spot auction held only 0.8 months before the delivery year began, compared to the approximately 37 months of the 2021/22 auction — the most recent held near the three-year forward horizon the RPM was designed around (Table 1). This compressed timeline eliminates any forward investment signal and makes the 2024/25 clearing price ($28.92/MW-day) an unreliable guide to equilibrium capacity incentives; the SFE framework, which embeds a long-run investment signal in the VRR demand curve, is not applicable.

# 6 Results: Equilibrium Prices vs. the Shapiro Settlement

## 6.1 Baseline Equilibrium

Table 4 presents the baseline symmetric SFE results for the three benchmark delivery years, calibrated with $K = 3$ strategic sellers and $c = \$150$/MW-day (the sector-average ACR for combined-cycle generators from the 2025 IMM State of the Market). For each year, the table reports the model-predicted equilibrium price $p^*$, the observed BRA clearing price, and the implied price-cost margin (Lerner index) $L = (p^* - c)/p^*$.

**Table 4:** Baseline SFE Equilibrium: $K = 3$, $c = \$150$/MW-day

| Year | Design | $\bar{p}$ | $\bar{p}_S$ | $p^*$ | $p_{\text{actual}}$ | $L$ | Outcome |
|---|---|---|---|---|---|---|---|
| 2023/24 | Old | \$412 | — | \$330 | \$34 | 0.55 | Interior |
| 2025/26 | Old | \$452 | — | \$426 | \$270 | 0.65 | Interior |
| 2026/27 | New | \$329 | \$325 | \$329 | \$329 | 0.54 | At cap |

*Notes:* $\bar{p}$ is the VRR Point (a) price cap. $\bar{p}_S$ is the Shapiro settlement cap (\$325/MW-day UCAP), applicable to the 2026/27 and 2027/28 delivery years; "—" indicates the settlement does not apply. $p^*$ is the model-predicted SFE clearing price. $p_{\text{actual}}$ is the observed BRA clearing price. $L$ is the price-cost margin (Lerner index) $(p^* - c)/p^*$. Prices in \$/MW-day. "Interior" means equilibrium falls strictly inside $(c, \bar{p})$; "At cap" means $S_i(\bar{p}) = \bar{q}_i$ for all $i$ and the Holmberg boundary condition is active.

The central finding of Table 4 for the settlement evaluation is that the Shapiro cap is *binding*: the SFE equilibrium price for 2026/27, $p^* = \$329$/MW-day, exceeds the settlement cap $\bar{p}_S = \$325$/MW-day. The gap is small at the RTO level (\$4/MW-day, roughly 1%), but it is structurally significant: it means that under the unmodified VRR curve and $K = 3$ strategic sellers, equilibrium prices would settle at the Point (a) cap rather than the lower settlement cap. The settlement displaces \$4/MW-day at the RTO level, but it does so by preventing the capacity-constrained boundary from operating, eliminating the price signal that would otherwise inform entry decisions. For constrained LDAs, the binding gap is larger: 2025/26 LDA prices reached \$444–\$466/MW-day, and the 2026/27 LDA equilibria likewise clear at or near $\bar{p}$ (see Section 6.5), implying settlement displacement of \$115–\$141/MW-day.

These binding gaps are evaluated further in Section 9 against the \$21 billion savings claim.

The 2026/27 result is the most informative for identification. The Holmberg boundary condition selects $p^* = \bar{p} = \$329.17$/MW-day for $K = 3$. It is important to be transparent about what this means: the matching of the observed clearing price is a *mechanical* consequence of the boundary condition, not an independent forecast. Once $K = 3$ is assumed and the 2026/27 capacity margin ($-0.2\%$) is tight enough to trigger the at-cap regime, the ODE solution is forced to $\bar{p}$ by construction. The dollar amount \$329.17 is simply the VRR Point (a) price read from the planning parameters. The substantively informative content is the *binary* at-cap versus interior outcome: for $K \geq 4$, the model predicts interior clearing strictly below the cap (see Section 6.2), so the observed at-cap outcome is consistent with $K \leq 3$ as the maximal effective seller count in 2026/27. The model does not confirm the price; it confirms the structural regime.

For 2023/24, the model predicts $p^* = \$330$ against an observed $p_{\text{actual}} = \$34$. The divergence is consistent with effective offer price mitigation. Under PJM's TPS rules, the three pivotal suppliers ($\text{RSI}_3 = 0.68$) had their offers capped at approximately the ACR level, and with an over-supplied market (capacity margin 9.9%), the clearing price fell to \$34. The SFE $p^* = \$330$ represents the *unmitigated* equilibrium — the price that would prevail if strategic sellers could bid without constraint. The 2025/26 gap ($p^* = \$426$ vs. $p_{\text{actual}} = \$270$) is smaller, reflecting the tighter market conditions (capacity margin 1.6%) and the price spike that occurred even under partial mitigation.

Figure 1 plots the equilibrium supply functions $s(p)$ and the VRR demand curves $D(p)$ for each benchmark year. The supply functions are solved by integrating the ODE (11) backward from the boundary condition at $\bar{p}$. All three supply functions are upward sloping in the interior of $(c, \bar{p})$, consistent with the SFE characterization. The equilibrium occurs where the aggregate supply $3 \cdot s(p^*)$ plus fringe supply meets the VRR demand.

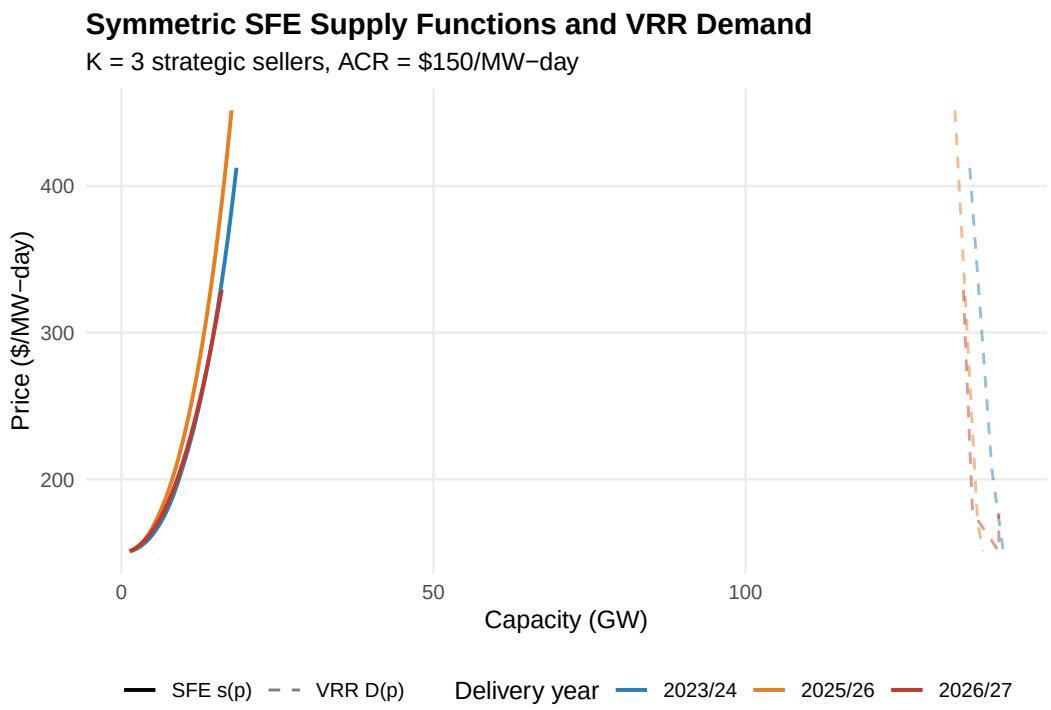


**Figure 1:** Symmetric SFE Supply Functions and VRR Demand Curves
*Notes:* Solid lines are equilibrium supply functions $s(p)$ for each benchmark year ($K = 3$, $c = \$150$/MW-day). Dashed lines are the corresponding VRR demand curves. Horizontal axis is GW; vertical axis is \$/MW-day.

## 6.2 Comparative Statics: Market Concentration

Figure 2 and Table 5 report the equilibrium price-cost margin $L$ as a function of the number of strategic sellers $K$, holding calibration-year parameters fixed. The results reveal three qualitative zones.

**Zone 1 ($K \leq 3$): High strategic markup, price cap binding.** For $K = 2$, the market clears at $\bar{p}$ in all three benchmark years, yielding Lerner indices of 0.64–0.67. The at-cap outcome is also produced by $K = 3$ in 2026/27 (where the VRR cap $\bar{p} = \$329$ is lower, meeting the equilibrium supply function sooner). The transition from $K = 2$ to $K = 3$ has a modest effect on the price-cost margin in 2023/24 (0.64 to 0.55) but almost none in 2025/26 (0.67 to 0.65), because in 2025/26 the supply function remains close to the cap throughout.

**Zone 2 ($K = 4$): Moderate strategic markup.** Adding a fourth seller reduces the price-cost margin by 33–45 percentage points relative to $K = 3$, shifting the 2023/24 equilibrium

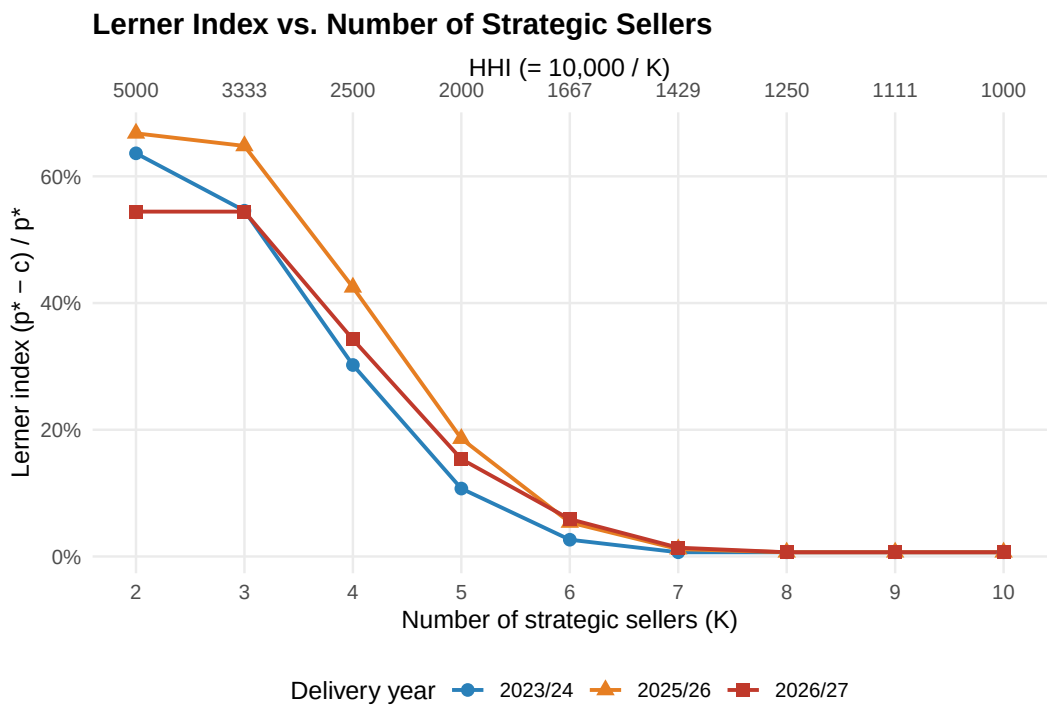


**Figure 2:** Price-Cost Margin vs. Number of Strategic Sellers
*Notes:* Each line traces the price-cost margin $L = (p^* - c)/p^*$ as $K$ varies from 2 to 10, holding all other calibration parameters fixed at their benchmark-year values. Shaded region indicates at-cap outcomes.

from \$330 to \$215 and the 2026/27 equilibrium from the cap to \$228. The ODE (11) makes this intuitive: increasing $K$ divides the denominator, forcing $s'(p)$ to be larger in absolute value at any given $(p, s)$ pair, which means firms must offer more capacity at every price below the cap and the equilibrium supply function shifts outward.

**Zone 3 ($K \geq 7$): Near-competitive.** With seven or more strategic sellers, equilibrium prices fall to within 0.5% of marginal cost ($c$ = \$150/MW-day) in all three years. The price-cost margin is effectively zero ($L < 0.02$), consistent with the competitive limiting result $p^* \to c$ as $K \to \infty$, implied by equation (11).

The structural implication is direct: the observed 2026/27 at-cap outcome ($p^*$ = \$329) is consistent with $K \leq 3$ but inconsistent with $K \geq 4$ (which would predict interior clearing at \$228 or below). This makes 2026/27 a useful natural experiment for establishing the effective number of strategic sellers.

**Table 5:** Equilibrium Price and Price-Cost Margin by $K$ (Baseline Calibration)

| | 2023/24 (Old VRR) | | | 2025/26 (Old VRR) | | | 2026/27 (New VRR) | | |
|---|---|---|---|---|---|---|---|---|---|
| $K$ | $p^*$ | $L$ | cap | $p^*$ | $L$ | cap | $p^*$ | $L$ | cap |
| 2 | 412 | 0.64 | ✓ | 452 | 0.67 | ✓ | 329 | 0.54 | ✓ |
| 3 | 330 | 0.55 | | 426 | 0.65 | | 329 | 0.54 | ✓ |
| 4 | 215 | 0.30 | | 261 | 0.42 | | 228 | 0.34 | |
| 5 | 168 | 0.11 | | 184 | 0.19 | | 177 | 0.15 | |
| 7 | 151 | 0.01 | | 152 | 0.01 | | 152 | 0.01 | |
| 10 | 151 | 0.01 | | 151 | 0.01 | | 151 | 0.01 | |

*Notes:* Prices in \$/MW-day. $L = (p^* - c)/p^*$; $c = \$150$/MW-day. ✓ indicates the market clears at the VRR price cap $\bar{p}$.

## 6.3 Comparative Statics: VRR Curve Design

Figure 3 shows the price-cost margin $L$ as a function of the VRR slope scaling factor. A factor of $f = 1$ corresponds to the baseline slope for each year; $f < 1$ means a flatter VRR (less elastic demand) and $f > 1$ a steeper VRR (more elastic demand on the sloped segment).

The results confirm the prediction from Equation (11): steeper VRR demand (larger $|D'(p)|$) reduces equilibrium markups by increasing the $D'(p)$ term in the ODE numerator, forcing a larger $s'(p)$ and thus a more competitive supply function. A doubling of the VRR slope reduces the Lerner index by roughly 20–30 percentage points in the interior-clearing benchmark years (2023/24 and 2025/26). For 2026/27 at $K = 3$, the outcome remains at the price cap until the slope factor exceeds approximately 1.4, at which point the supply function becomes steep enough to clear below the cap.

The 2026 VRR redesign (FERC Order ER25-1357) steepened the sloped segment between $p_f$ and $\bar{p}$ by reducing the quantity width of that segment, consistent with a slope scaling factor slightly above 1.0. In isolation, this steepening would reduce markups. However, the new design also introduces a flat floor at $p_f \approx \$177$/MW-day, which truncates the lower tail of the price distribution and prevents equilibrium prices from falling below $p_f$ regardless of $K$. This floor effect dominates at high $K$: with $K = 5$ and the new design, the minimum clearing price is \$177 (the floor), compared to \$168 with the old design (which allows clearing down to $c = \$150$).

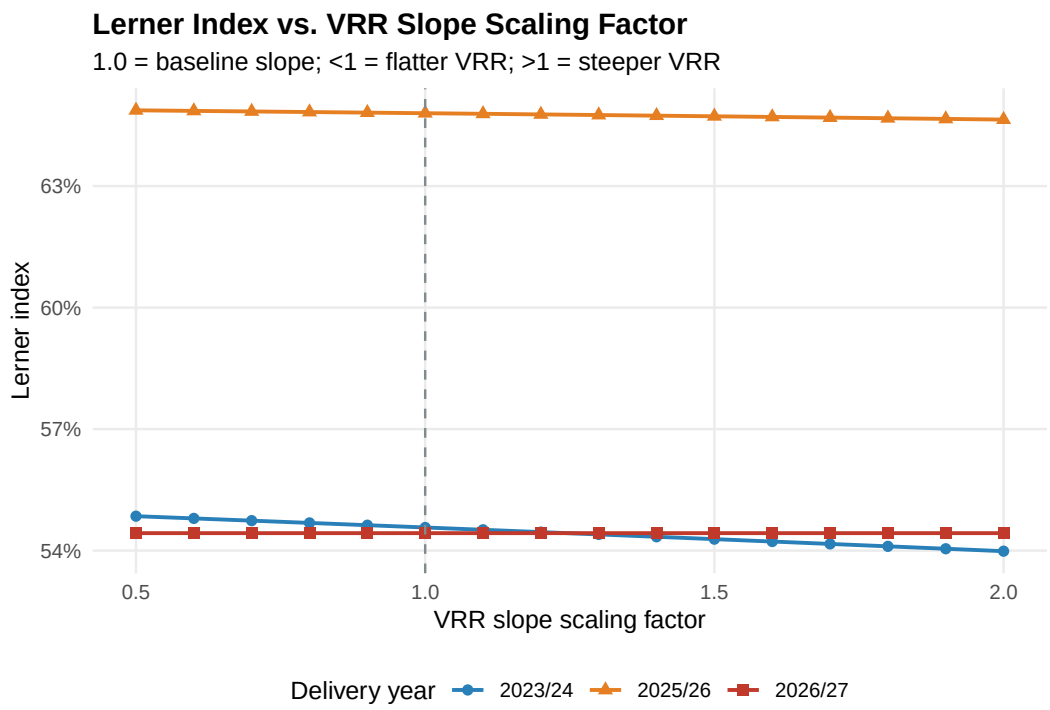


**Figure 3:** Price-Cost Margin vs. VRR Slope Scaling Factor ($K = 3$)
*Notes:* The slope scaling factor multiplies the quantity increment ($q_b - q_a$) on the sloped segment, leaving the anchor prices fixed. Factor = 1 is the calibrated baseline.

## 6.4 Fringe Supply and Transmission Counterfactual

Figure 4 traces the price-cost margin $L$ as the fringe supply $\bar{Q}_f$ varies from 70% to 150% of its calibrated baseline, holding all other parameters fixed. The fringe captures competitive imports and non-strategic capacity, so this comparative static approximates the effect of expanding the Capacity Emergency Transfer Limit (CETL) or reducing transmission congestion.

Increasing fringe supply by 30% (factor = 1.3) reduces the price-cost margin by 12–18 percentage points in the interior-clearing years, corresponding to equilibrium price reductions of \$30–\$75/MW-day. The mechanism is straightforward: more fringe capacity shifts the market-clearing condition (12) rightward, requiring the strategic sellers to clear at a lower point on their supply functions where the markup is smaller.

Notably, for 2026/27, expanding the fringe at $K = 3$ reduces the price-cost margin from

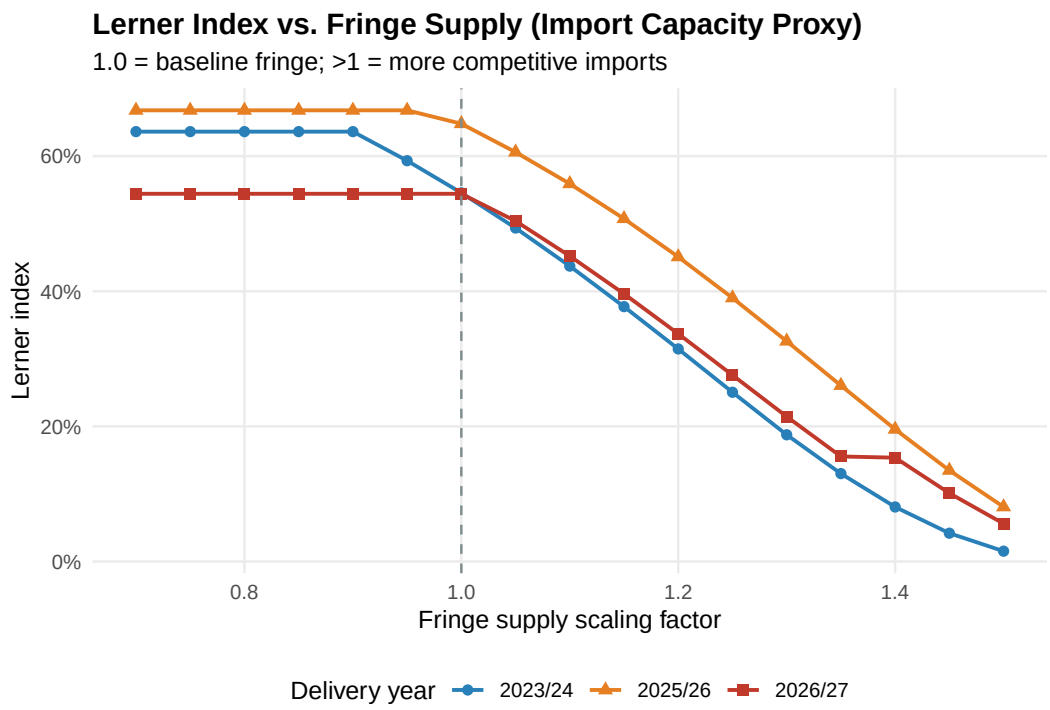


**Figure 4:** Price-Cost Margin vs. Fringe Supply Scaling Factor ($K = 3$)
*Notes:* Fringe supply scaling factor of 1.0 equals the calibrated $\bar{Q}_f$ for each benchmark year. Fringe bids at $c = c_f$ (same ACR floor). The fringe supply step function shifts the equilibrium clearing quantity without affecting the ODE itself.

0.54 to roughly 0.30 at factor = 1.3, which corresponds to a price reduction from the cap ($329) to approximately $215/MW-day. This suggests that expanding import capacity to the RTO would have a substantial effect on equilibrium pricing even without structural remedies to seller concentration. The implied cost-effectiveness of transmission investment versus divestiture depends on the marginal cost per MW of CETL capacity — a comparison we leave for future work.

## 6.5 LDA-Level Market Power

The RTO-level analysis treats PJM as a single market, but capacity is traded separately in constrained Locational Deliverability Areas. For each LDA, the CETL imposes a hard upper bound on competitive imports, and local strategic sellers cover the residual reliability requirement. This section applies the same SFE model to each LDA individually, calibrating

the competitive fringe at $\bar{Q}_f$ = CETL and setting each strategic seller's capacity at $\bar{q}$ = (rel. req. − CETL)/$K$.

Table 6 reports the implied Lerner indices for all LDAs with valid CETL data across the three benchmark years. Two patterns dominate.

**Table 6:** SFE Lerner Index by LDA and Benchmark Year ($K = 3$, $c = \$150$/MW-day)

| LDA | 2023/24 | 2025/26 | 2026/27 |
|---|---|---|---|
| ATSI | 0.62$^\dagger$ | 0.63 | 0.54$^\dagger$ |
| ATSI-Cleveland | 0.61 | 0.56 | 0.54$^\dagger$ |
| COMED | 0.65$^\dagger$ | 0.67$^\dagger$ | 0.54$^\dagger$ |
| DEOK | 0.60$^\dagger$ | 0.48 | 0.54$^\dagger$ |
| DOM | – | 0.66 | 0.54$^\dagger$ |
| DPL SOUTH | 0.62$^\dagger$ | 0.67 | 0.54$^\dagger$ |
| EMAAC | 0.66$^\dagger$ | 0.68$^\dagger$ | 0.54$^\dagger$ |
| JCPL | – | – | 0.54$^\dagger$ |
| MAAC | 0.64$^\dagger$ | 0.67$^\dagger$ | 0.54$^\dagger$ |
| PEPCO | 0.63$^\dagger$ | – | 0.54$^\dagger$ |
| PL | 0.64$^\dagger$ | 0.65 | 0.54$^\dagger$ |
| PS | 0.67$^\dagger$ | 0.70$^\dagger$ | 0.54$^\dagger$ |
| PS NORTH | 0.67$^\dagger$ | 0.70$^\dagger$ | 0.54$^\dagger$ |
| SWMAAC | 0.59$^\dagger$ | 0.68$^\dagger$ | 0.54$^\dagger$ |
| BGE | 0.55$^\dagger$ | 0.68 | 0.54 |

*Notes:* $K = 3$ strategic sellers per LDA. $c = \$150$/MW-day (CC technology-average ACR, 2025 IMM SotM). Market structure calibrated using $\bar{Q}_f$ = CETL (competitive imports) and $\bar{q}$ = (rel. req. − CETL)/$K$ (strategic sellers cover local gap). $^\dagger$ Market clears at or within \$0.50 of the VRR price cap.

**Uniformly high price-cost margins.** Lerner indices range from 0.48 (DEOK in 2025/26) to 0.70 (PS and PS NORTH in 2025/26), with most LDAs above 0.60 in every year. The elevated price-cost margins are consistent with low import penetration ratios: across the sample, CETL/rel. req. averages 0.40, meaning strategic sellers must cover 60% of the reliability requirement on average. In 2026/27, every LDA clears at or within \$0.50 of the \$329.17 price cap — perfectly consistent with the RTO result and confirming that the at-cap outcome is not an artifact of RTO aggregation.

**Transmission constraints amplify price-cost margins.** Figure 6 illustrates the relationship between import penetration and the price-cost margin. The dashed trend lines (estimated separately per year) are uniformly downward-sloping: LDAs with higher CETL

relative to their reliability requirement systematically exhibit lower equilibrium markups. A 10-percentage-point increase in the import penetration ratio is associated with a 4–6-point reduction in the price-cost margin depending on the benchmark year. DEOK, which has the highest import penetration ratio in the sample at 96% (2025/26), is the only LDA to produce an interior equilibrium price below the RTO price ($290 vs. $270).

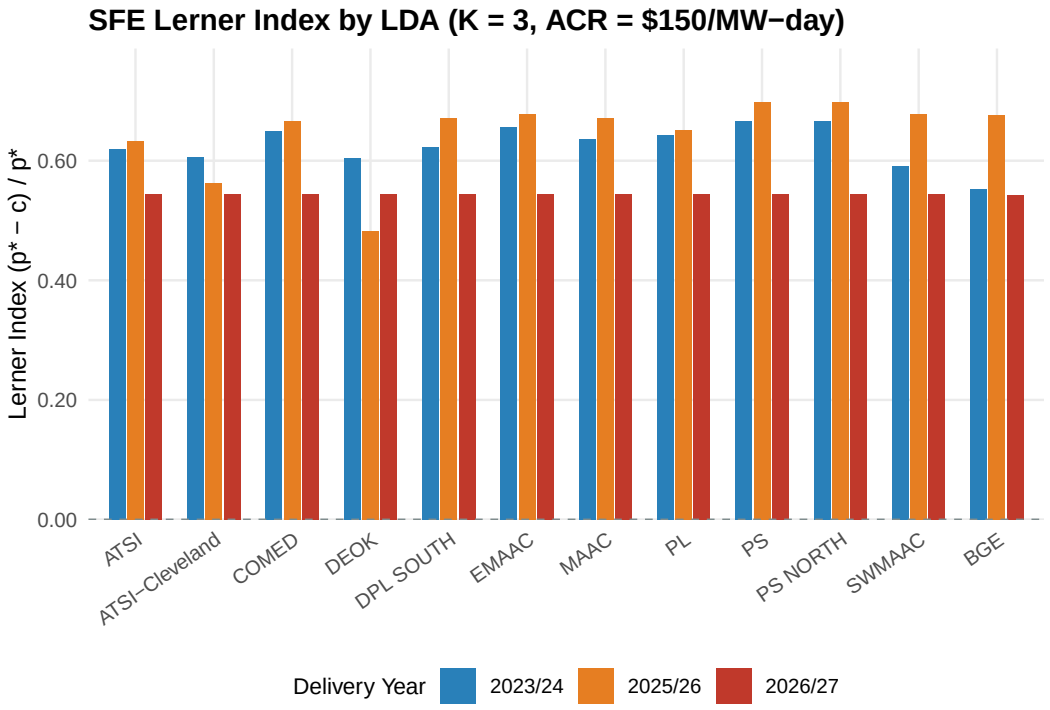


**Figure 5:** SFE Price-Cost Margin by LDA and Benchmark Year ($K = 3$)
*Notes:* Bars show the SFE price-cost margin $L = (p^* - c)/p^*$ for each LDA and benchmark year. $K = 3$, $c = \$150$/MW-day. LDAs ordered by 2026/27 price-cost margin. † indicates the market cleared at or within $0.50 of the VRR price cap.

These results have a direct policy implication. The CETL is the primary instrument available to PJM for reducing local strategic pricing without structural divestiture. The cross-LDA variation confirms the comparative static from Section 6.4: expanding the CETL is quantitatively as important as reducing strategic concentration for LDAs in the middle of the import-penetration distribution. For LDAs at the left tail — PS, PS NORTH, COMED, and EMAAC — even large CETL expansions would leave price-cost margins above 0.50 given the current number of strategic sellers, indicating that structural remedies are necessary to

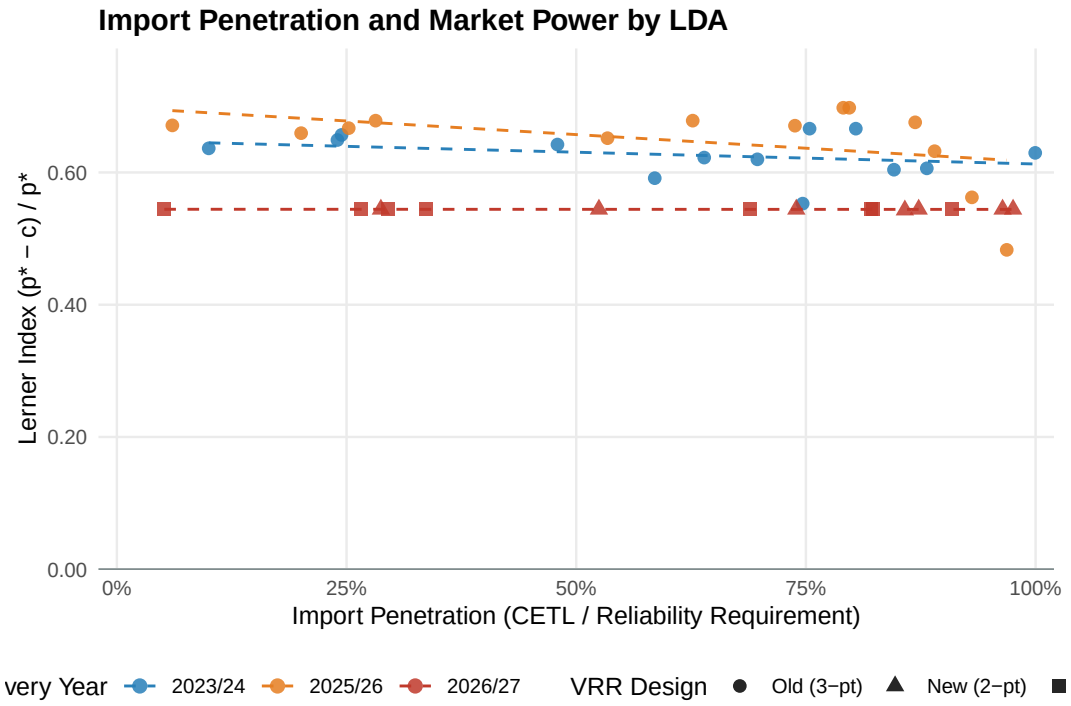


**Figure 6:** Import Penetration and Price-Cost Margin by LDA
*Notes:* Horizontal axis is CETL/reliability requirement, the share of local demand that can be served by competitive imports. Vertical axis is the SFE price-cost margin $L$. Each point is one LDA-year observation. Dashed lines are OLS trends estimated separately per year. Shapes distinguish VRR design variants.

achieve competitive outcomes.

# 7 The Allocative Cost of Price Controls

The Shapiro settlement imposes a hard cap at $\bar{p}_S = \$325$/MW-day UCAP on PJM's capacity demand curve for the 2026/27 and 2027/28 delivery years. Section 6 established that the SFE equilibrium $p^* = \$329$ exceeds $\bar{p}_S$ in 2026/27 — the settlement is binding by \$4/MW-day at the RTO level and by \$115–\$141/MW-day in 2025/26 constrained LDAs. Binding a price cap below the equilibrium changes market outcomes through two channels: a revenue transfer from capacity sellers to buyers, and a distortion of the entry and exit incentives that the capacity price is designed to convey. This section analyzes both channels. Section 9 quantifies the revenue transfer in relation to the \$21 billion savings claim.

## 7.1 Revenue Transfer and the Net CONE Benchmark

Table 7 summarizes the key price levels and the implied revenue transfer per MW-year under the settlement. For the 2026/27 delivery year, the SFE equilibrium falls at the VRR Point (a) price $\bar{p} = \$329.17$/MW-day, yielding a binding gap of \$4.17/MW-day relative to the settlement cap. At 365 days per year, this corresponds to a revenue transfer of approximately \$1,522 per MW-year at the RTO level. For 2027/28, the VRR Point (a) price under the new design is \$333.44/MW-day, implying a binding gap of \$8.44/MW-day and a revenue transfer of approximately \$3,081 per MW-year.

**Table 7:** Settlement Price Levels vs. SFE Equilibrium and Net CONE (2026/27 and 2027/28)

| Year | $\bar{p}$ | $\bar{p}_S$ | $p^*$ | Net CONE | $\bar{p}_f$ | $p^* - \bar{p}_S$ |
|---|---|---|---|---|---|---|
| 2026/27 | \$329.17 | \$325.00 | \$329.17 | \$212.14 | \$175.00 | \$4.17 |
| 2027/28 | \$333.44 | \$325.00 | \$333.44 | \$242.52 | \$179.55 | \$8.44 |

*Notes:* All prices in \$/MW-day UCAP. $\bar{p}$ is the VRR Point (a) price cap; $\bar{p}_S = \$325$/MW-day is the Shapiro settlement cap; $p^* = \bar{p}$ because the SFE clears at the cap for $K \leq 3$ (both years observed at-cap in BRA data); Net CONE is the published Net Cost of New Entry for the RTO zone (Bresler and Keech, 2025); $\bar{p}_f$ is the settlement floor (Point (b)/(c) in the new VRR curve); $p^* - \bar{p}_S$ is the binding gap. Revenue transfer per MW-year $= (p^* - \bar{p}_S) \times 365$.

The revenue transfer from sellers to buyers per MW-year is modest relative to the settlement's stated savings — \$1,522 and \$3,081 per MW-year in 2026/27 and 2027/28 respectively, compared to the \$21 billion aggregate claim, which uses a counterfactual price of over \$500/MW-day and a much larger binding gap. Section 9 reconciles these estimates. The relevant comparison here is not the size of the transfer per MW-year, but its proximity to the Net CONE investment threshold. In 2026/27, Net CONE is \$212.14/MW-day. The settlement cap at \$325/MW-day is well above Net CONE (ratio 1.53), so in isolation the cap does not prevent new entry from being financially viable: a generator earning \$325/MW-day exceeds its Net CONE by 53%. The investment distortion from the cap alone is therefore limited in 2026/27.

## 7.2 Entry Distortion and the Informational Role of Prices

The more substantive allocative concern is not whether prices remain above Net CONE today, but whether suppressing the price signal degrades the forward-looking information that investors use to evaluate entry decisions for the 2028/29 delivery year and beyond. Hayek (1945) argues that prices perform an indispensable coordinative function: they aggregate and transmit dispersed private knowledge about scarcity conditions that no central authority can replicate. In a capacity market, the scarcity signal is the gap between the clearing price and Net CONE. A price of \$329/MW-day signals acute scarcity (\$117/MW-day above Net CONE in 2026/27); capping it at \$325/MW-day preserves the directional signal but compresses the magnitude.

The informational loss is larger when the cap is binding in consecutive years. If the 2027/28 auction also clears at the settlement cap (\$325 with $\bar{p} = \$333.44$ and Net CONE = \$242.52), potential entrants observe only the cap rather than the unconstrained equilibrium price. They cannot distinguish between "the market would have cleared at \$333, signaling \$91/MW-day above Net CONE" and "the market is capped and might have cleared anywhere above \$325." This uncertainty reduces the informativeness of the capacity price for investment decisions, a concern also raised in the price-caps-in-SFE literature (Anderson and Xu, 2005).

The settlement's applicability to constrained LDAs amplifies this effect. In 2026/27, all LDA-level SFE equilibria fall at or near the respective VRR Point (a) caps (Section 6.5), but the binding gap between $p^*$ and $\bar{p}_S$ is structurally larger in tightly constrained LDAs because those LDAs face steeper demand curves and higher Net CONE values (e.g., PS and PS NORTH Net CONE = \$369/MW-day in 2026/27, far above both $\bar{p}_S$ and $\bar{p}$). In such LDAs, the settlement cap truncates a price signal that was already struggling to attract investment.

## 7.3 The Floor's Distinct Allocative Role

The \$175/MW-day floor is not the mirror image of the cap. It operates as an investment adequacy mechanism: all capacity offered at or below \$175/MW-day UCAP is procured at the floor price, regardless of whether the market would have cleared there (Bresler and Keech, 2025). PJM characterizes the floor explicitly as "supporting near-term investment" — it prevents the price from collapsing in over-supplied market conditions, protecting the revenue stream for marginal capacity.

A natural benchmark for evaluating the floor is Net CONE. If the floor exceeds Net CONE, it provides a revenue guarantee above the investment threshold, functioning as an implicit subsidy to existing capacity that may not be needed for reliability. At the RTO level, the 2026/27 floor (\$177.24/MW-day in VRR parameters) is *below* Net CONE (\$212.14/MW-day): the floor does not guarantee full cost recovery and thus does not over-compensate existing generators on net. In the SWMAAC zone, however, the relationship inverts: SWMAAC Net CONE is \$171.02/MW-day while the floor is \$177.24/MW-day, so the floor exceeds the local investment threshold by \$6.22/MW-day and provides above-cost revenue to SWMAAC capacity.

The floor's interaction with the cap creates a price band: outcomes between \$175 and \$325 are interior equilibria; outcomes outside this band are truncated. For $K = 3$, the SFE equilibrium in 2026/27 falls at \$329 — above the band. The settlement cap therefore does not allow the capacity price to reach the strategic equilibrium; it instead pins the outcome to the top of the band. From the perspective of the standard missing-money literature (Cramton and Stoft, 2005; Joskow, 2008), a price floor is a design-level intervention to correct the missing money problem in energy-only markets. Applying a floor simultaneously with a cap creates an internally inconsistent incentive: the cap signals that scarcity is not severe enough to warrant prices above \$325, while the floor signals that prices below \$175 are inadequate to support investment. The two signals are not contradictions only if the competitive equilibrium falls in the interior of the band — which it does not under the $K = 3$ structural regime the

SFE model identifies.

### 7.4 The Price Band as a Non-Competitive Equilibrium

Glaeser and Luttmer (2003) demonstrate in the context of residential rent control that binding price controls generate misallocation beyond the standard deadweight triangle: when quantity is rationed below the market- clearing level, allocation is distorted away from highest-willingness-to-pay consumers. The direct mechanism — relationship-based allocation in rental housing — does not translate to capacity markets, where all capacity offering at or below the cap clears. The relevant analogue for capacity markets is inter-temporal: a binding cap suppresses the price signal that should direct investment toward future delivery years with the highest expected scarcity, creating an efficiency loss that accumulates dynamically rather than appearing as a static misallocation of current capacity.

The settlement band ($175–$325) identifies a price interval that is neither the competitive equilibrium ($c \approx$ $150, far below the floor) nor the strategic equilibrium ($p^* \approx$ $329–$333, above the cap). It is an administratively imposed outcome. Whether this intervention is welfare-improving depends on whether the strategic markup — of order $4–$8/MW-day above the cap — is large enough to justify the associated information loss and the investment distortion from cap-binding. The SFE model suggests the markup is modest (1–3% above the cap) at the RTO level, supporting the view that the cap's revenue transfer benefit is small relative to its informational cost. Section 10 considers whether structural remedies can reduce $p^*$ itself, rather than truncating the equilibrium price signal.

## 8 Evaluating the Lead-Time Argument

### 8.1 The Complaint's Investment-Adequacy Theory

The legal theory underlying the Shapiro complaint is not a market-conduct claim. Pennsylvania did not allege that capacity sellers withheld supply strategically or bid above competitive

levels. The complaint alleged instead that PJM's VRR curve was "miscalibrated" — that its price cap was "too high given lower expectations for new entry in the near-term" (Bresler and Keech, 2025). The implicit causal chain is: if new generation cannot be financed, permitted, and constructed before the delivery year begins, then high capacity prices cannot stimulate entry within the relevant horizon, and elevated prices serve only as a transfer from buyers to sellers rather than as an investment signal. On this view, the demand curve should reflect what consumers are willing to pay for *available* capacity, not what would be needed to attract capacity that cannot arrive in time.

This is an investment-adequacy argument about auction design, not a finding of strategic conduct. It is worth separating the two because the SFE model speaks directly to the second and only indirectly to the first. The model confirms that the 2026/27 at-cap outcome is consistent with strategic pricing by $K \leq 3$ sellers; it does not address whether new entry could have responded to those prices within the available lead time. The lead-time argument is therefore independent of the SFE findings — it is a claim about the supply side's medium-run elasticity, not about the current-period strategic equilibrium.

## 8.2 PJM's Lead-Time Record and the Settlement Horizon

Section 3.4 documented the extreme compression of BRA lead times from 2021/22 through 2024/25. The 2024/25 BRA was held in May 2024 — just 0.8 months before the June 2024 start of the delivery year — making it effectively a spot procurement with no meaningful entry window. This compression is the empirical foundation for the lead-time argument: high prices in a spot auction cannot plausibly attract new entry, so the price cap's calibration to long-run entry costs was disconnected from actual market conditions.

However, the settlement applies to the 2026/27 and 2027/28 delivery years, for which the auction schedule had been substantially restored. The 2026/27 BRA was held in July 2025, approximately 10.2 months before the June 2026 delivery start; the 2027/28 BRA was held in January 2026, approximately 16.8 months ahead (Bresler and Keech, 2025). These

lead times remain short relative to the 36–48 months required to finance and construct a new combined-cycle gas plant, but they substantially exceed the 0.8-month spot horizon that motivates the complaint's sharpest form.

For demand response, battery storage, and capacity upgrades to existing plants — resource types with deployment horizons of 6–18 months — a 10.2-month lead time is non-trivial. The empirical question is whether the 2026/27 price spike ($269.92/MW-day in the 2025/26 BRA; the 2026/27 BRA cleared at $329.17) triggered supply-side responses in these faster-moving categories before the auction. PJM's Capacity Interconnection Queue and Demand Response registration data would be needed to answer this directly, which is beyond the scope of the current analysis. The point is that the lead-time argument is strongest for greenfield generation and weaker for incremental and demand-side resources, and these distinctions matter for evaluating whether the complaint's causal claim holds quantitatively.

## 8.3 Five Channels: Why Prices Still Convey Information

Even if new greenfield entry is infeasible within a 10-month window, auction prices serve at least five informational and allocative functions beyond directly stimulating construction:

1. **Retirement deterrence.** High prices reduce the probability that marginal capacity retires before or during the delivery period. At $329/MW-day, a 100 MW generator earns approximately $12 million per year from capacity revenue alone — well above the threshold that would typically trigger a retirement decision. Capping the price at $325/MW-day preserves this incentive nearly intact (the cap is only $4/MW-day binding at the RTO level), but the precedent of administrative price suppression increases uncertainty about future clearing prices, potentially accelerating retirement decisions in subsequent years.

2. **Mobile resource deployment.** Demand response programs, behind- the-meter storage, and contracted interruptible loads can register for capacity auctions on timescales

of weeks to months. High prices provide the financial incentive to activate these registrations. The settlement cap reduces the annual payment for a 1 MW DR registration from \$120,085 (at $p^* = \$329$/MW-day) to \$118,625 (at $\bar{p}_S = \$325$/MW-day) — a difference of \$1,460/MW-year that is small in absolute terms. The more consequential signal degradation is forward-looking: if investors anticipate that future auctions will similarly be capped, the present value of the registration's upside is truncated for resources evaluating participation in the 2028/29 auction and beyond.

3. **Portfolio contracting and hedging.** Load-serving entities and generation owners enter bilateral capacity contracts in advance of BRAs. Observed BRA prices inform the forward strip that underlies these contracts. A cap that suppresses BRA prices therefore also suppresses the contract prices that hedge investment risk, increasing the cost of capital for prospective developers regardless of lead time.

4. **Regulatory and interconnection signaling.** Sustained high capacity prices signal to state regulators, transmission planners, and interconnection queue managers that the region is capacity-constrained. This signal prompts regulatory approvals, permitting, and transmission investments that improve the supply environment for future auctions. Capping the price shortens the signal's duration and reduces its magnitude.

5. **Expectations for future periods.** Investors planning capacity for 2028/29 and beyond observe 2026/27 and 2027/28 BRA outcomes to form expectations about future price levels. If the settlement binds in both years, the observed prices (\$325) understate the strategic equilibrium (\$329–\$333), leading to under-investment relative to a world where prices reflected unconstrained scarcity. Cramton and Stoft (2005) emphasize that the capacity market's primary function is to provide a long-run investment signal; a two-year cap disrupts this function precisely in the years when the signal is most informative.

## 8.4 The Hayek Critique and the Correct Remedy

Hayek (1945) argues that the central epistemological problem of economic organization is the utilization of knowledge that is not given to any single mind in its totality. The price system solves this problem by aggregating dispersed information — about costs, preferences, and scarcity — into a summary statistic that enables decentralized coordination. Price controls suppress this statistic regardless of whether the underlying information would otherwise have triggered an immediate supply response. A price cap at \$325/MW-day destroys the signal that $p^* = \$329$ even if no generator can break ground on a new plant within 10 months.

The appropriate policy response to the lead-time problem is to restore auction timing, not to suppress auction prices. Weitzman (1974) analyzes the choice between price and quantity regulatory instruments under uncertainty, showing the optimal instrument depends on the relative slopes of marginal benefit and cost. In the capacity market context, the VRR curve's explicit downward-sloping demand design embeds the regulator's preference for a price instrument over a fixed quantity mandate — an instrument choice that is more informationally efficient when the equilibrium price location is uncertain. Suppressing that price with a cap undermines the instrument the market design was built to use. Restoring a 36-month lead time allows prices to elicit entry before they must be suppressed; capping prices without restoring lead time removes the instrument before it can function.

PJM's own settlement framing is consistent with this priority ordering. The settlement slides describe the 2026/27 BRA (July 2025) and 2027/28 BRA (January 2026) as a partial restoration of the forward-market schedule (Bresler and Keech, 2025). The settlement is presented as a temporary compromise for the two auctions whose schedules were most disrupted by the 2021–2025 irregularities, not as a permanent recalibration of the VRR curve. The temporary nature of the settlement — two delivery years, after which the tariff would revert to the original VRR calibration or a reformed version — limits the cumulative informational damage but does not eliminate it.

## 8.5 Summary

The lead-time argument provides a coherent justification for treating the 2026/27 and 2027/28 settlements as emergency measures during a period of acute auction-schedule disruption. Its empirical strength depends on the deployment horizon of the marginal resource: the argument is strongest for greenfield gas generation (36–48 months) and weakest for demand response, storage, and existing-plant upgrades (6–18 months). For the 2026/27 BRA held 10.2 months ahead, prices could have attracted resources in the second category, and the informational channels enumerated in Section 8.3 remain active regardless of lead time. The correct structural remedy for the lead-time problem is schedule restoration — which PJM is pursuing — rather than price suppression, which eliminates the scarcity signal that a restored schedule would otherwise transmit.

# 9 Evaluating the $21 Billion Claim

## 9.1 The Counterfactual Behind the Claim

Pennsylvania's January 28, 2025 announcement stated that the settlement would save consumers "over $21 billion over the next two delivery years" (Office of Governor Josh Shapiro, 2025). The arithmetic behind this figure is straightforward: (counterfactual price − $325) × cleared quantity × 2 delivery years. Shapiro's stated counterfactual is "over $500/MW-Day" (Bresler and Keech, 2025). This figure represents the projected VRR Point (a) price for 2026/27 under the *pre-settlement* calibration methodology, which relied on the Gross CONE of the dual-fuel combustion turbine reference technology before netting out projected energy and ancillary service revenues. Under the pre-reform methodology, a higher Gross CONE would have produced a Point (a) substantially above the $329.17 produced by the new-design calibration. The 2025/26 Point (a) of $451.61 (Table 3) is consistent with this range; the complaint projected a further increase to "over $500" for 2026/27 absent reform.

This counterfactual deserves scrutiny. The VRR Point (a) price is not a strategic equilibrium price. It is the maximum of the administrative demand schedule: the price at which the VRR curve becomes perfectly inelastic (the "flat top" in the old design). In equilibrium, the capacity market clears where supply meets the VRR schedule — which, as Section 6 shows, may occur anywhere in the range $[c, \bar{p}]$ depending on market structure. For $K = 3$, the model predicts interior equilibria in 2023/24 and 2025/26 that fall substantially below Point (a). The VRR counterfactual therefore conflates two separate components of a high clearing price: the scarcity rent that is *part of the market design* (the gap between the competitive price $c$ and the Point (a) cap $\bar{p}$) and the strategic markup above competitive levels (captured by the SFE price-cost margin $L$).

Our SFE model provides a structural alternative counterfactual: $p^*$, the equilibrium price under strategic bidding against the *unmodified* VRR curve. This is what the market would have cleared at absent the settlement, given the observed market structure. In 2026/27, $p^* = \$329.17$/MW-day; in 2027/28, $p^* = \$333.44$/MW-day (the observed BRA clearing price, consistent with the Holmberg at-cap condition; Bresler and Keech 2025). Both equilibria fall at the respective VRR Point (a) caps because the Holmberg boundary condition is active for $K \leq 3$, but they are substantially below the \$500 figure in Shapiro's claim.

## 9.2 Three-Scenario Revenue-Transfer Analysis

Table 8 compares the implied revenue transfer under three counterfactual scenarios for the 2026/27 and 2027/28 delivery years, using RTO-level cleared quantities from the published BRA results.

Several observations follow from Table 8.

**The VRR counterfactual overstates the settlement's direct benefit.** Using \$500/MW-day as the counterfactual and RTO-level quantities, the two-year revenue transfer is \$17.16 billion — approximately 18% below the stated \$21 billion. The gap arises primarily from constrained LDAs, where the settlement cap applies to the separately-clearing LDA

**Table 8:** Revenue Transfer Under Three Counterfactual Scenarios (RTO Level)

| | Counterfactual price | | Annual revenue transfer | | Two-year |
|---|---|---|---|---|---|
| Scenario | 2026/27 | 2027/28 | 2026/27 | 2027/28 | total |
| VRR Point (a) [Shapiro] | \$500 | \$500 | \$8.57B | \$8.59B | \$17.16B |
| SFE equilibrium [this paper] | \$329 | \$333 | \$0.20B | \$0.41B | \$0.61B |
| Competitive equilibrium | \$150 | \$150 | (cap exceeds competitive; transfer reversed) | | |
| Settlement cap | \$325 | \$325 | — | — | — |

*Notes:* All prices in \$/MW-day UCAP. Annual revenue transfer = $(p_{\text{counterfactual}} - \$325) \times Q \times 365$, where $Q = 134{,}205$ MW (2026/27) and $Q = 134{,}478$ MW (2027/28) from BRA clearing data. VRR Point (a) counterfactual uses \$500/MW-day as a conservative approximation of Shapiro's "over \$500" figure; the two-year RTO total of \$17.16B falls short of the \$21B claim, with the remainder attributable to constrained LDA effects (see text). SFE counterfactual uses model-predicted $p^* = \bar{p}$ for $K \leq 3$. Competitive equilibrium ($c = \$150$/MW-day) implies the settlement cap transfers revenue *to* sellers relative to the competitive outcome.

demand curves. In 2025/26, PJM's constrained LDA prices reached \$444–\$466/MW-day; without the settlement, 2026/27 and 2027/28 LDA equilibria in the tightest zones (PS, PS NORTH, EMAAC) would likely have cleared near their respective VRR caps as well, contributing additional revenue transfer not captured in the RTO-only calculation.

**The SFE counterfactual yields a far smaller transfer.** The structural estimate based on $p^*$ implies a two-year revenue transfer of approximately \$612 million — 35 times smaller than the \$21 billion VRR-based claim. This does not mean the settlement is inconsequential: the SFE and VRR counterfactuals differ not because one is wrong, but because they answer different questions. The VRR counterfactual measures the gap between the design-level cap and the settlement cap — a policy decision about where to set the demand curve ceiling. The SFE counterfactual measures the gap between the strategic equilibrium and the settlement cap — the reduction in strategic markup attributable to the price band. The first gap (\$175/MW-day) is much larger than the second (\$4–\$8/MW-day) because the strategic equilibrium in 2026/27 happens to fall near the VRR Point (a) price.

**The competitive benchmark inverts the direction of transfer.** If capacity markets achieved the competitive equilibrium ($c \approx \$150$/MW-day), the settlement cap at \$325 would represent a transfer *to* sellers, not buyers — \$175/MW-day above marginal cost. This is unsurprising: capacity markets are *designed* to clear above marginal cost to provide investment

signals (Cramton and Stoft, 2005; Joskow, 2008). The relevant benchmark for evaluating Shapiro's savings claim is not competitive pricing but rather the strategic equilibrium, which is the price that would prevail in the market as structured.

### 9.3 Quantity Assumption and Supply Response

All three scenarios in Table 8 hold quantity fixed at the observed BRA clearing. This assumption is internally consistent for short-run analysis — the capacity that cleared in 2026/27 and 2027/28 was the supply available given the lead times involved — but it suppresses supply response. PJM's settlement framing acknowledges this implicitly: the $175 floor is described as "supporting near-term investment," recognizing that capping the price without a floor could accelerate retirements and reduce future clearing quantities.

A complete savings calculation should account for the effect of the price cap on (i) retirement decisions for marginal capacity that would have been retained under higher prices, and (ii) new entry decisions for the 2028/29 auction and beyond, which are informed by the observed 2026/27 and 2027/28 clearing prices. Both effects reduce future cleared quantities, reducing the denominator of the cost-per-MW calculation. The $21 billion figure holds quantity fixed, so it is an upper bound on consumer savings even under the VRR counterfactual. The SFE estimate of $612 million is both a lower counterfactual and a smaller quantity-adjusted figure, making it an approximate lower bound on the settlement's direct price effect.

## 10 Policy Alternatives

The analysis in Sections 7–9 establishes that the Shapiro settlement reduces the equilibrium capacity price by $4–$8/MW-day at the RTO level, suppresses the scarcity signal that informs entry decisions, and generates aggregate consumer savings that are an order of magnitude smaller than the VRR-counterfactual estimate of $21 billion. This section considers four

structural alternatives that address the source of elevated capacity prices — a concentrated seller side with limited competitive fringe — rather than suppressing the price symptom. The alternatives are not mutually exclusive; the most durable solution likely combines elements of all four.

## 10.1 Option 1: The Shapiro Settlement (Current Policy)

The settlement reshapes the VRR demand curve for 2026/27 and 2027/28, imposing a cap at $\bar{p}_S = \$325$/MW-day UCAP and a floor at $\bar{p}_f = \$175$/MW-day UCAP. It is designed as a temporary emergency measure during the period of acute auction-schedule disruption documented in Section 3.4, and PJM's slides describe it as applicable to two specific delivery years only (Bresler and Keech, 2025).

The settlement's key advantages are speed and certainty: the VRR curve modification is directly within FERC's regulatory authority under FPA §205, it does not require structural changes to the seller side, and it provides immediate price relief for the 2026/27 and 2027/28 auctions. Its key disadvantages are equally clear. It does not reduce the number of strategic sellers, expand the competitive fringe, or steepen the VRR demand curve; it therefore does nothing to address the structural source of the $K \leq 3$ regime. Section 7 shows that the price-cost margins in 2026/27 would remain at $L = 0.54$ under the settlement — unchanged from the SFE equilibrium. The settlement compresses the *price*, but not the underlying *markup structure*.

Furthermore, as Section 8 argues, suppressing the price signal imposes an informational cost that persists beyond the settlement window: investors planning for 2028/29 and later years observe capped prices rather than the unconstrained equilibrium, reducing the clarity of the capacity market's investment signal.

## 10.2 Option 2: Transmission Expansion (CETL)

The comparative statics in Section 6.4 show that increasing the competitive fringe supply $\bar{Q}_f$ reduces equilibrium prices organically, without requiring any modification to the demand curve. Fringe supply in the SFE model represents capacity that bids at (or near) marginal cost, which in PJM's context corresponds to imports limited by the Capacity Emergency Transfer Limit. A 30% increase in the competitive fringe at $K = 3$ reduces the price-cost margin from 0.54 to approximately 0.30 in 2026/27 — a reduction of 24 percentage points, far larger than the settlement's zero-point-zero reduction in price-cost margins.

The LDA cross-sectional results (Section 6.5) confirm this mechanism: a 10-percentage-point increase in the import penetration ratio CETL/reliability requirement is associated with a 4–6-point reduction in the LDA price-cost margin. For the median LDA in the sample (import penetration 0.40), increasing the CETL to achieve a 0.50 import penetration ratio would reduce the price-cost margin by roughly 6–12 points, equivalent to a price reduction of \$20–\$40/MW-day at the 2026/27 VRR cap.

CETL expansion through transmission investment preserves the price signal: equilibrium prices fall because competition increases, not because the demand curve is truncated. The scarcity signal remains informative; it is simply responding to a market structure with more competitive imports. This makes transmission expansion the most structurally efficient of the four options considered here. The relevant policy instrument is PJM's Regional Transmission Expansion Plan (RTEP), which can be directed to prioritize projects that increase import capacity to capacity-constrained LDAs.

## 10.3 Option 3: VRR Curve Redesign

Section 6.3 shows that steepening the VRR demand curve — increasing $|D'(p)|$ on the sloped segment — reduces equilibrium markups through the ODE mechanism. The intuition is straightforward: a steeper demand curve means each unit of capacity withheld has less effect on price (the curve is harder to move), which reduces the marginal incentive to withhold.

A doubling of the VRR slope reduces the price-cost margin by 20–30 percentage points in interior-clearing years; for 2026/27, the at-cap outcome transitions to interior clearing at a slope scaling factor of approximately 1.4.

FERC Order ER25-1357 redesigned the VRR curve for the 2026/27 BRA, steepening the sloped segment between $p_f$ and $\bar{p}$ by narrowing its quantity width. This is a step in the right direction from an SFE perspective. However, the redesign also introduced the flat floor at $p_f \approx \$177$/MW-day, which as Section 7.3 discusses, creates an investment-adequacy guarantee whose interaction with the cap introduces its own distortions.

A reformed VRR curve that steepens the slope further while maintaining a smooth lower tail (without a hard floor) would more directly address the strategic markup problem. The quantitative target from the SFE results is a slope factor of approximately 1.4 or greater: at that steepness, the 2026/27 equilibrium transitions from cap-binding to interior-clearing even with $K = 3$ sellers, allowing the price signal to operate freely below the Point (a) cap.

## 10.4 Option 4: Entry Promotion and Deconcentration

The $K$ comparative static in Section 6.2 is the most direct guide to durable structural reform. The transition from cap-binding to interior-clearing equilibria occurs between $K = 3$ and $K = 4$: adding a fourth symmetric strategic seller reduces the price-cost margin by 20–33 percentage points and moves the 2026/27 equilibrium from \$329 to \$228 (a reduction of \$101/MW-day). With $K \geq 7$, equilibrium prices fall to within 0.5% of marginal cost in all three benchmark years.

Reducing effective seller concentration in PJM's capacity market requires lowering the barriers that prevent entry and expansion. Three levers are available:

1. **Interconnection queue reform.** PJM's interconnection queue has historically taken 4–7 years to process new generator applications, far exceeding the lead times at which high capacity prices can stimulate entry. FERC Order 2023 on transmission planning and Order 2023-A on interconnection reform are designed to accelerate this process.

Timely interconnection is a necessary condition for the $K \geq 4$ outcome that would organically resolve the cap-binding equilibrium.

2. **Demand response and storage deployment.** Demand response resources and utility-scale battery storage can participate in PJM's capacity market on timescales much shorter than new thermal generation. Expanding the effective capacity of these resources increases $K$ in the SFE framework by adding sellers who do not share the symmetric cost structure of the thermal strategic sellers. This is an asymmetric effect: demand response that bids at zero marginal cost is analogous to increasing the competitive fringe rather than increasing $K$, providing the fringe-expansion benefits described in Option 2.

3. **Targeted divestiture or mitigation.** For the LDAs at the left tail of the import-penetration distribution — PS, PS NORTH, COMED, and EMAAC — even large CETL expansions leave price-cost margins above 0.50 given the current seller count (Section 6.5). In these areas, structural remedies including divestiture or enhanced offer price mitigation may be necessary to achieve interior clearing. PJM's existing Capacity Market Seller Offer Cap mitigation rules provide a framework for this, though they have historically been calibrated to the Avoidable Cost Rate rather than the SFE-based markup threshold.

## 10.5 Comparative Assessment

Table 9 summarizes the four options along the dimensions that matter for capacity market design: the source of price reduction, the duration of the effect, whether the price signal is preserved, and the required regulatory authority.

Under the model's assumptions, Options 2–4 dominate Option 1 along every structural dimension: they produce permanent price reductions by addressing the source of high prices (concentrated supply, limited competitive fringe, inelastic demand) rather than the symptom

**Table 9:** Comparative Assessment of Policy Alternatives

| Option | Price reduction mechanism | Duration | Signal preserved | Regulatory lever |
|---|---|---|---|---|
| 1. Settlement (current) | Cap/floor reshape VRR | 2 years | No (cap binding) | FERC §205 tariff |
| 2. CETL expansion | Competitive fringe ↑ | Permanent | Yes | RTEP, FERC Tx planning |
| 3. VRR redesign | Demand elasticity ↑ | Permanent | Yes | FERC VRR recalibration |
| 4. Entry/deconcentration | $K \uparrow$ / fringe ↑ | Permanent | Yes | FERC Ords. 2023/2023-A; mitigation |

*Notes:* Signal preservation indicates whether the capacity clearing price reflects unconstrained strategic equilibrium (Options 2–4) or is truncated by an administrative cap (Option 1). Duration refers to whether the price reduction persists beyond a fixed delivery window.

(high clearing prices). A regulator facing uncertainty about market structure — for instance, whether $K = 2$ rather than $K = 3$ — might rationally prefer the settlement's certain near-term price reduction while structural reforms are pursued. The settlement is justified as an emergency measure during a period of acute schedule disruption and lead-time compression, but it should not substitute for the structural reforms that would render a similar intervention unnecessary in future auctions. PJM's own settlement framing calls for "reform the capacity market" alongside the cap (Bresler and Keech, 2025), consistent with this ranking.

# 11 Conclusion

On December 30, 2024, Pennsylvania filed a FERC complaint arguing that PJM's capacity auction demand curve was miscalibrated given near-term entry expectations. One month later, Governor Shapiro announced a settlement: the VRR curve for the 2026/27 and 2027/28 auctions would be reshaped, with a cap at \$325/MW-day UCAP and a floor at \$175/MW-day UCAP, at a projected savings of over \$21 billion for 65 million consumers. This paper uses a calibrated Supply Function Equilibrium model to evaluate the settlement's terms, its stated counterfactual, and its consequences for entry incentives and information.

Five findings emerge.

**First, the settlement is binding under the observed market structure.** With $K \leq 3$ pivotal suppliers — $\text{RSI}_3 < 1$ in every benchmark year establishes an upper bound, and the TPS test showing three individually pivotal sellers supports $K = 3$ over $K = 2$ — the SFE equilibrium for 2026/27 falls at the VRR Point (a) cap of \$329.17/MW-day, exceeding the settlement cap $\bar{p}_S = \$325$ by \$4.17/MW-day at the RTO level. The at-cap outcome is consistent with $K \leq 3$: four or more symmetric strategic sellers would predict interior clearing at \$228 or below (Section 6.2). In constrained LDAs, where the 2025/26 BRA produced prices of \$444–\$466/MW-day, the structural gap between the SFE equilibrium and the settlement cap is considerably larger. The settlement reduces the price paid in the 2026/27 and 2027/28 auctions; it does not reduce the price-cost margin $L$, which remains 0.54 under the strategic equilibrium.

**Second, the settlement modifies the demand curve, not merely the price outcome.** The settlement does not impose an external overlay; it replaces the VRR curve shape with a flat top at \$325, the existing sloped segment between \$325 and \$175, and a flat bottom at \$175 that procures all sub-threshold supply at the floor price. The \$175 floor is an investment-adequacy mechanism — below the RTO Net CONE of \$212/MW-day in 2026/27 — designed to prevent the price from collapsing in over-supplied future auctions. Applying a floor while capping the price creates an administered price band that is neither the competitive equilibrium (\$150/MW-day) nor the strategic equilibrium (\$329–\$333/MW-day). The incentive implications of the band depend on where the unconstrained equilibrium falls relative to the Net CONE threshold, which varies across LDAs (Section 7.3).

**Third, the \$21 billion savings figure uses the wrong counterfactual.** Shapiro's claim is computed as the gap between the unconstrained VRR Point (a) price ("over \$500/MW-day") and the settlement cap, multiplied by cleared quantity over two delivery years. Our SFE model provides a structural alternative: the strategic equilibrium $p^*$ against the unmodified VRR curve. Using RTO-level quantities, the VRR counterfactual implies a two-year transfer

of approximately \$17.16 billion at the RTO level (constrained-LDA effects account for the gap to the \$21 billion headline, though the precise LDA contribution requires LDA-level cleared quantity data not analyzed here); the SFE counterfactual implies a two-year transfer of approximately \$612 million (Section 9). The two estimates answer different questions. The VRR-based figure measures the gap between the design ceiling and the settlement cap — a policy decision about how to calibrate the demand curve. The SFE-based figure measures the reduction in strategic markup attributable to the price band. The first is 35 times larger than the second because the 2026/27 strategic equilibrium happens to fall near the design ceiling, not because the settlement eliminates large strategic rents.

**Fourth, the lead-time argument for the settlement is partial.** The complaint's investment-adequacy theory — that the VRR cap is too high given near-term entry expectations — is strongest for greenfield generation (36–48 month deployment) and weakest for demand response, storage, and existing-plant upgrades (6–18 months). For the 2026/27 BRA held 10.2 months ahead, at least some capacity categories could have responded to price signals. More fundamentally, prices inform investment even when direct construction is infeasible within the lead time: through retirement deterrence, mobile resource deployment, portfolio contracting, regulatory signaling, and the formation of expectations for future delivery years (Section 8.3). The correct structural remedy for lead-time compression is schedule restoration, which PJM is pursuing, not price suppression.

**Fifth, structural alternatives dominate price controls.** Three interventions reduce SFE equilibrium prices without suppressing the scarcity signal: CETL expansion (30% fringe increase $\rightarrow -24$ pp price-cost margin reduction; 10 pp CETL/reliability-requirement increase $\rightarrow$ 4–6 pp reduction); VRR slope steepening beyond the ER25-1357 baseline (slope factor $\geq 1.4$ moves 2026/27 from cap-binding to interior clearing even at $K = 3$); and entry promotion to reach $K \geq 4$ (which alone would reduce 2026/27 prices by \$101/MW-day without any demand-curve modification). Each of these options is permanent and preserves the price signal; the settlement is temporary and suppresses it (Section 10).

The central finding is that PJM's 2026/27 and 2027/28 capacity prices are high because the market is structurally thin — three sellers hold pivotal capacity in every benchmark year — and because transmission constraints limit competitive imports in the most constrained LDAs. The settlement addresses the symptom (high prices) rather than the cause (concentrated supply with limited competitive fringe). It is defensible as emergency relief during a period of acute auction-schedule disruption; it is not a substitute for the transmission investment, VRR redesign, and entry promotion that would make a similar intervention unnecessary in future auctions.

Several directions for future work follow naturally from the model's limitations. An asymmetric-firm SFE would characterize how cost and capacity heterogeneity interact with the TPS screen and the settlement's LDA application. A dynamic model with endogenous entry and exit would connect the short-run price-cost margin findings to long-run capacity adequacy under the price band. An empirical structural model recovering firm-level cost parameters from observed offer data would allow direct welfare calculation and would permit testing whether the $K = 3$ symmetric calibration understates or overstates actual strategic markups. The calibrated SFE results in this paper provide a structural benchmark for each of these extensions and a quantitative foundation for evaluating the settlement's long-run consequences as the 2028/29 auction cycle approaches.

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